

Frontier Vision-Language Models: Architectural Evolution, Benchmarks, Applications, and Challenges

Zongxia Li* Xiyang Wu* Hongyang Du
Fuxiao Liu Huy Nghiem Guangyao Shi

{zli12321, wuxiyang, hdu1, fl3es, nghiemh}@umd.edu
shig@usc.edu

<https://github.com/zli12321/Vision-Language-Models-Overview.git>

Abstract

Large vision-language models (VLMs) have evolved rapidly from contrastive image-text encoders and adapter-based assistants into natively multimodal foundation models that support long-context reasoning, agentic workflows, and, increasingly, unified perception-generation-action loops. Since 2025, the frontier has consolidated around a small number of large pretraining families, including GPT, Gemini, Claude, Grok, Qwen, Gemma, DeepSeek, Kimi, and MiniMax, while evaluation has shifted from short-form visual question answering toward spatial, temporal, embodied, and calibration-sensitive benchmarks. In this survey, we provide an updated overview of VLMs through 2026 with an emphasis on three developments. First, we trace the architectural evolution from contrastive or bridged two-tower models to LLM-backbone adapter models, native multimodal-input models, omni-modal unified input/output models, and emerging world-action models. Second, we summarize how benchmarks expanded from classical OCR, VQA, and chart understanding toward long-video reasoning, embodied evaluation, reward-model judging, and domain-specific decision making. Third, we review the growing role of post-training, including supervised fine-tuning, preference optimization, and reinforcement learning methods such as GRPO-style multimodal alignment. Rather than exhaustively listing every release, we focus on representative frontier families, influential benchmarks, and the major open challenges that remain in hallucination, safety, efficiency, data quality, and multimodal alignment.

1. Introduction

Pretrained large language models (LLMs), such as LLaMA [206] and GPT-4 [169], have achieved remarkable success across a wide range of NLP tasks [145, 156]. As these systems continue to scale [160], they face two persistent constraints: the finite supply of high-quality text data [118, 211] and the inherent limitation of single-modality models for tasks that require joint visual, spatial, temporal, and linguistic reasoning [56, 74]. These pressures motivate the development of large vision-language models (VLMs), which combine visual inputs such as images and videos with text to produce richer representations of objects, scenes, events, and abstract concepts [21, 65]. VLMs already power visual question answering, web and GUI agents, document understanding, robotics, and autonomous driving [2, 205]. At the same time, they introduce failure modes that are weaker in text-only systems, including visual hallucination, brittle grounding, and shallow multimodal reasoning [60, 130]. The field changed especially quickly in 2025–2026. Frontier systems are no longer isolated point releases, but fast-moving multimodal families from a small number of labs: GPT [170], Gemini [41], Claude [8], Grok [230], Qwen [39], Gemma [57], DeepSeek [228], Kimi [4], and MiniMax [153]. Across these families, the center of gravity has shifted from attaching a vision encoder to a text model toward pretraining a single multimodal trunk with longer context, stronger tool use, and better support for agentic workflows. In this paper, we provide a critical examination of modern VLMs with three emphases: the architectural evolution of frontier systems, the expansion of evaluation beyond classical VQA-style tests, and the growing role of post-training and reinforcement learning in multimodal reasoning. Rather than attempting an exhaustive live index of every release, we fo-

cus on representative frontier families and benchmark directions that clarify how the field is changing.

2. State-of-the-Art VLMs

Recent VLM progress is best understood as the evolution of a small number of pretraining families rather than a flat list of independent models. Figure 1 highlights how public model scale grew from sub-billion contrastive systems to frontier multimodal families, while Table 1 summarizes representative brand-name releases that define the current landscape.

Two shifts stand out from Table 1. First, the dominant units of progress are now families, each with multiple reasoning modes, context tiers, and deployment targets. Second, architectural progress is no longer well described by the simple story of “add a vision encoder to an LLM.” The frontier moved from contrastive or bridged two-tower systems such as CLIP [183] and Flamingo [5], to LLM-backbone adapter models, and then to native multimodal pretraining where image, video, audio, and text share a single trunk.

We use this architectural view throughout the rest of the paper. Section 3 expands the building blocks behind each era. Section 4 examines how evaluation evolved from short-form answer matching toward spatial, temporal, embodied, and calibration-aware assessment. Section 5 then connects these capabilities to computer-use agents, document intelligence, media generation, and embodied systems. Finally, the later sections discuss why post-training, alignment, safety, and data quality have become central bottlenecks as frontier VLM families grow more agentic and multimodal.

3. Building Blocks and Training Methods

The evolution of VLMs is no longer captured by a single transition from “train from scratch” to “attach vision to an LLM.” As Figure 2 and Table 1 show, the field now spans several architectural eras: contrastive or bridged two-tower systems, projector-based LLM-backbone models, native multimodal-input models, unified input/output models, and emerging world-action models. Across these eras, the same ingredients still recur: a visual tokenizer or encoder, a language-side tokenizer and decoder, alignment modules or shared token spaces, and increasingly sophisticated post-training. What changed is where modality fusion happens, whether outputs remain text-only, and whether the model is expected merely to describe the world or also to generate and act within it. In this section, we summarize the most foundational architectural components of VLMs and then explain how frontier systems reorganize them across eras. We conclude with the recent post-training shift from supervised instruction tuning toward preference optimization and

reinforcement learning for multimodal reasoning.

3.1. Common Architecture Components

Vision Encoder plays a crucial role in projecting visual components into embedding features that align with embeddings from large language models (LLMs) for tasks such as text or image generation [48]. It is trained to extract rich visual features from image or video data, enabling integration with language representations [143, 260].

Specifically, vision encoders used in many VLMs [34, 40, 129, 214], are pretrained on large-scale multimodal or image data: These encoders are jointly trained on image-text pairs, allowing them to capture visual and language relationships effectively. Notable examples include CLIP [183], which aligns images and text embeddings via contrastive learning, and BLIP [113], which leverages bootstrapped pretraining for robust language-image alignment. Pretrained on large scale ImageNet [43] or Similar Datasets: These encoders are trained on vast amounts of labeled visual data or through self-supervised training [173], enabling them to capture domain-specific visual features. While initially unimodal, these encoders, such as ResNet [67] or Vision Transformers (ViTs) [46], can be adapted for multimodal tasks. They excel at extracting meaningful object-level features and serve as a solid foundation for vision-language models. Many SoTA VLMs, such as Qwen2-VL [214] and LLaVA [129], commonly incorporate pretrained vision encoders. These encoders not only provide robust and meaningful visual representations but are also highly effective for transfer learning [251]. They outperform randomly initialized encoders [73] by leveraging learned vision knowledge from their training domains.

Text Encoder projects tokenized text sequences into an embedding space, similar to how vision encoders process images. Models such as CLIP [183], BLIP [113], and ALIGN [90] use both an image encoder and a text encoder. These models use contrastive learning to align image and text embeddings in a shared latent space, effectively capturing cross-modal relationships. However, newer models, such as LLaVA [129], often do not include a dedicated text encoder. Instead, they rely on large language models (LLMs) (e.g., LLaMA [206], Vicuna [37]) for text understanding, integrating visual inputs through projection layers or cross-attention mechanisms [124]. This shift shows a growing trend of using the capabilities of LLMs over vision components for more versatile and advanced multimodal reasoning and generation tasks.

Text Decoder leverages LLMs as the primary text generator, using visual encoders to project image features [100]. GPT-4V [169], Flamingo [5], and Kosmos-2 [175] use this approach. These models typically use a minimal visual projection mechanism, allowing the powerful language decoder to generate contextually rich outputs. ViLBERT and Visual-

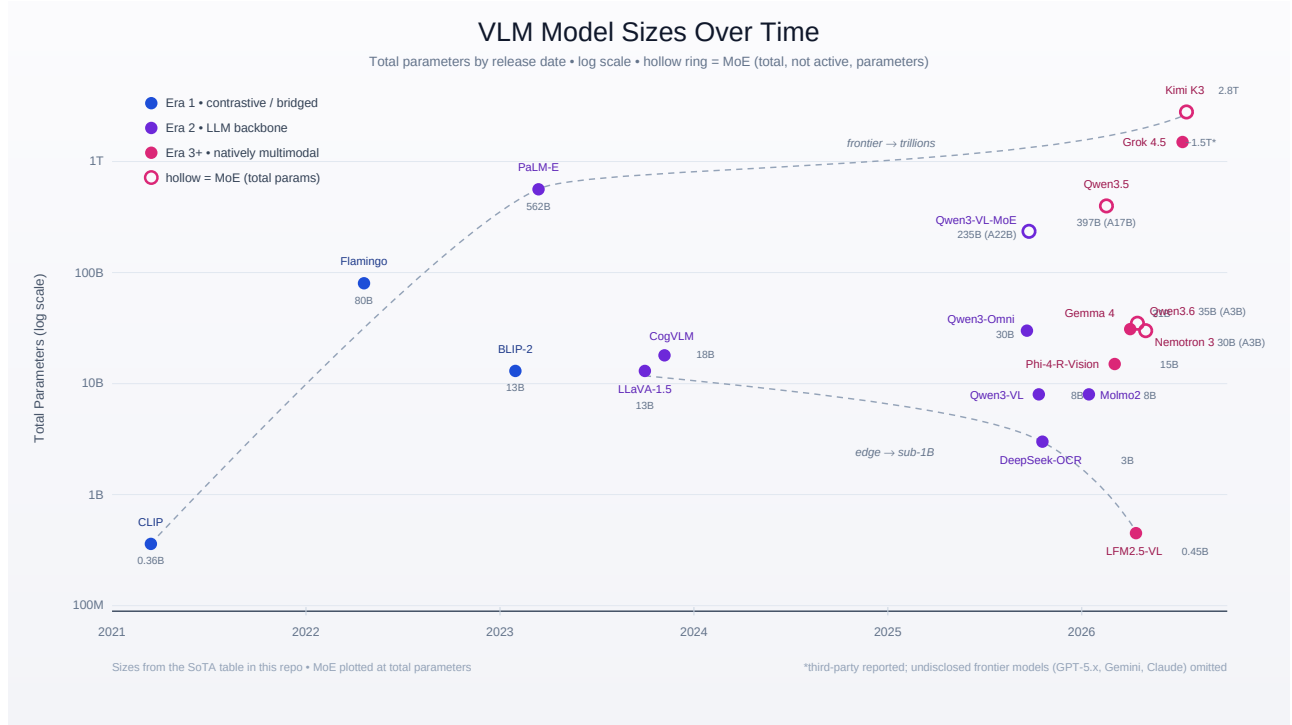


Figure 1. Selective timeline of representative VLM families. Hollow markers denote mixture-of-experts models plotted by total rather than active parameters. Frontier proprietary families with undisclosed parameter counts, including GPT, Gemini, and Claude, are discussed in the text and Table 1 but cannot be placed exactly on the curve.

Family	Representative releases	Era	Public scale	Snapshot
OpenAI	GPT-4V [240], GPT-5.6 Sol [170]	2 → 3a	Undisclosed	Closed frontier family for multi-modal reasoning, coding, and computer use
Google DeepMind	Gemini [6], Gemini 3.6 Flash [41]	2 → 3a	Undisclosed	Workhorse multimodal family with 1M-context agentic reasoning
Anthropic	Claude 3 [7], Claude Opus 4.6 [8]	2 → 3a	Undisclosed	Long-context multimodal family focused on code and agentic workflows
xAI	Grok 4.5 [230, 231]	3a	Reported ~1.5T	Vision-capable coding and agentic model with 500K context
Alibaba Qwen	Qwen-VL [12], Qwen2.5-VL [13], Qwen3.6-35B-A3B [182], Qwen3.8-Max [39]	2 → 3a/3b	2.4T / 95B active	Family now spans compact open checkpoints and a 2.4T multimodal flagship
Google Gemma	Gemma 4 [57]	3a	31B dense; 26B MoE (3.8B active)	Compact open multimodal family built from the same research as Gemini 3
DeepSeek	DeepSeek-VL2 [228], DeepSeek-OCR-2 [221]	2	4.5B active (VL2)	Open multimodal family with strong OCR, document, and vision-language reasoning
Moonshot Kimi	Kimi K3 [4]	3a	2.8T / 104B active	Open 3T-class native multimodal flagship with 1M context
MiniMax	MiniMax M3 [153]	3a	428B / 23B active	Native multimodal 1M-context model for coding, video, and computer use

Table 1. Selective frontier VLM families through 2026. We emphasize major pretraining families from large labs rather than attempting to list every derived checkpoint or task-specific open model.

The Evolution of Vision-Language Model Architectures

contrastive / bridged two-towers → LLM backbones → natively-fused inputs → omni-modal I/O → world-action models

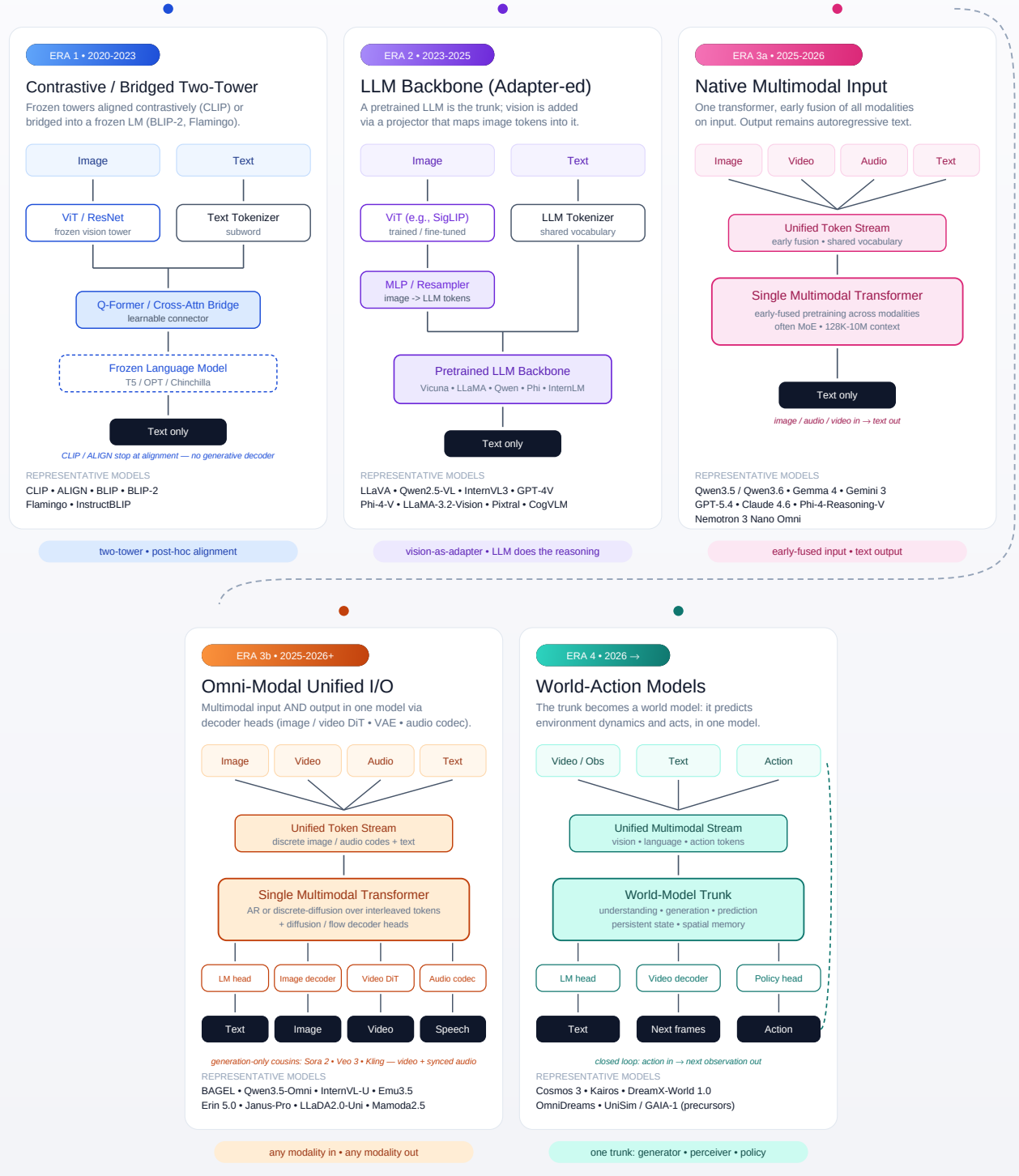


Figure 2. Architectural evolution of modern VLMS. Era 1 aligns separate vision and language towers contrastively or through a learned bridge. Era 2 makes a pretrained LLM the trunk and injects visual tokens through a projector or resampler. Era 3a early-fuses multimodal inputs into a single transformer but still emits text. Era 3b extends the same trunk with image, video, and audio decoder heads for unified input/output generation. Era 4 adds action and persistent state, yielding emerging world-action models that act as generator, perceiver, and policy in one network.

BERT [115, 137] provide the foundation to decoder architectures for multimodal pretraining. Training VLMs from scratch typically requires a separate text decoder, whereas using LLMs as the backbone often uses the original decoders from the LLM. (Figure 2).

Cross-Attention Mechanisms enable visual-text interactions by allowing tokens from one modality (vision) to influence tokens from the other modality (text) [124]. These layers compute attention scores across modalities, but not all models use them. ViLBERT [137] and Flamingo [5] employ cross-attention, while CLIP [183] does not.

3.2. Building Blocks of Training From Scratch

Training a VLM from scratch typically uses distinct training objectives and methodologies compared to using an LLM as the backbone. Self-Supervised Learning (SSL) pre-trains without needing human labeled data to scale up pretraining [68]. Variants of SSL techniques include masked image modeling [69], contrastive learning [209], and image transformation prediction [154]. In this section, we delve into contrastive learning, a common pre-training process to scale up VLM training from scratch.

Contrastive Learning employs separate encoders for visual and textual inputs, mapping them into a shared embedding space. The visual encoder extracts features using convolutional neural networks (CNN) [171] or vision transformers (ViTs) [46]. The text encoder processes textual inputs into embeddings. Contrastive learning aligns related image-text pairs by minimizing the distance between their visual and text embeddings in the shared space, while maximizing the distance between embeddings of unrelated pairs. Pioneering models like CLIP [183], BLIP [113], and ALIGN [90] leverage this approach, pre-training on large-scale image-text datasets to develop robust, transferable representations for downstream tasks.

3.3. Building Blocks of Using LLMs as Backbone

Large Language Models serve as the text generation component that processes encoded visual and textual inputs to produce text outputs autoregressively [25, 169, 206] for VLMs. In the context of VLMs, LLMs include their original text decoders. In this section, we list two common ways to align visual and pre-trained LLM text features.

Projector maps visual features extracted by the vision encoder into a shared embedding space aligned with the text embeddings from the LLM. It typically consists of multi-layer perceptron (MLP) layers [159], which transform high-dimensional visual representations into compact embedding tokens compatible with the textual modality. The projector can be trained jointly with the rest of the model to optimize cross-modal objectives or freezing certain parts of the model, such as the LLM, to preserve pre-trained knowledge. Projectors are the defining connector of Era 2

systems. Representative examples include LLaVA [129], Qwen2-VL [214], DeepSeek-VL2 [228], and Pixtral [3]. In newer native multimodal families, the explicit projector becomes smaller, more distributed, or disappears altogether because all modalities are pretrained into a single trunk.

Joint Training is an end-to-end approach that updates weights of all components of the model in parallel without freezing any weights, including the LLM and projector layers. This approach has been used in models such as Flamingo [5].

Freeze Training Stages involves selectively freezing model components during training, preserving pre-trained knowledge while adapting to new tasks [76]. Common strategies include freezing pre-trained vision encoders while fine-tuning projector layers, and implementing gradual unfreezing of components [176] or freezing LLM layers while only updating vision encoder weights [207].

3.4. Newer Architectures

Recent frontier work is less about inventing new standalone components than about changing where fusion happens and what the multimodal trunk is expected to output.

Era 2: projector-based LLM backbones. In adapter-era systems, the language model remains the reasoning core and visual features are injected through a projector, resampler, or cross-attention bridge. GPT-4V [240], LLaVA [129], Qwen2-VL [214], DeepSeek-VL2 [228], and Pixtral [3] exemplify this recipe. The design is modular and efficient for instruction tuning, but it can leave vision as a secondary pathway rather than a fully shared representational space.

Era 3a: native multimodal input, text output. The strongest 2025–2026 frontier families move beyond adapter-style fusion and instead pretrain a single multimodal trunk that directly consumes text, images, documents, audio, and video. In this regime, modalities are aligned earlier and more uniformly, while outputs remain primarily autoregressive text. Representative families include Gemini 3.6 Flash [41], Claude Opus 4.6 [8], GPT-5.6 Sol [170], Grok 4.5 [230], Qwen3.8-Max [39], Kimi K3 [4], MiniMax M3 [153], and Gemma 4 [57]. This design supports much longer context windows and makes multimodal perception central rather than auxiliary.

Era 3b: unified multimodal input/output. Some families push further and ask the same trunk not only to understand multiple modalities, but also to generate them. Emu3 [217], Janus-Pro [33], and related unified models treat image, video, audio, and text as parts of one token or latent interface, typically with decoder heads such as VAE-, DiT-, or codec-based modules. The practical motivation is to collapse understanding and generation into one backbone rather than coupling separate specialist models.

Era 4: world-action models. The newest step extends the

multimodal trunk into a predictive control model. World-action systems such as Cosmos 3 [165], Kairos [204], DreamX-World 1.0 [203], and OmniDreams [166] treat action as another modality and optimize the model to predict future observations while selecting actions. In other words, the model becomes generator, perceiver, and policy in a single loop. Even when these systems are not yet the dominant general-purpose assistant architecture, they reveal where multimodal scaling is heading.

3.5. VLM Alignments

Alignment now affects not only safety and helpfulness, but also whether frontier VLMs can sustain long reasoning traces over visual evidence. The broad recipe remains inherited from LLM alignment: supervised fine-tuning defines formats and tool use, preference optimization improves response quality and safety, and reinforcement learning increasingly targets hard multimodal reasoning skills. Direct Preference Optimization (DPO) [184] and Proximal Policy Optimization (PPO) [190] remain important foundations, but the strongest 2025–2026 trend is the adoption of GRPO-style rule-based reinforcement learning for visual reasoning, spatial reasoning, and multimodal generation.

While RLHF succeeds on LLMs, VLMs add at least two new alignment burdens: the model must reason over evidence that is only partially linguistic, and the reward signal must distinguish true visual grounding from plausible but unsupported verbal answers. When a model handles image inputs alongside text, it can reveal or infer sensitive details, over-trust its language prior, or misread crucial visual context. These concerns make multimodal alignment harder than merely porting text-only recipes. Recent work therefore mixes several signals: human preference pairs [259], rule-based correctness rewards [83, 152, 256], constrained rewards for spatial faithfulness [96], and teacher or verifier guidance that avoids the full cost of dense human labeling [81, 104, 236]. The broader lesson is that post-training is now one of the main levers separating a capable multimodal base model from a practically useful reasoning model. The key to RLHF is to collect human feedback and design reward models. In [259], authors introduce a high-quality, human-annotated dataset with 120k preference comparison pairs to enhance the alignment of VLMs. It proposes a Critique-Based Reward Model that improves interpretability by generating critiques before assigning scores. By contrast, [152, 264] extends large-scale rule-based reinforcement learning to multimodal scenarios and reproduces key characteristics of text-based RL (like DeepSeek-R1 [62]) in visual contexts. Despite using only simple, sparse rewards (format and accuracy) and minimal data filtering, the authors achieve stable improvements in accuracy and response length.

In addition to RLHF, Reinforcement Learning with Ver-

ifiable Rewards (RLVR) is also getting attention [136]. RLVR bypasses the need for training a reward model by utilizing a direct verification function to evaluate correctness. This method streamlines the reward process while ensuring strong alignment with the task’s correctness criteria.

4. Benchmarks and Evaluation

The number of VLM benchmarks has grown rapidly with the quick development of new VLMs since 2022 [36, 255]. Comprehensive benchmarking is important for evaluating model performance and ensuring robust training across different capabilities various aspects such as math reasoning, scene recognition, etc [58, 138]. Modern VLM benchmarks have moved far beyond basic VQA. The most recent frontier benchmarks test whether a model can track a scene over time, act through an embodied interface, maintain calibration on document extraction, or even serve as a reliable judge for long-horizon agents [51]. In this section, we summarize representative benchmark categories rather than commit to a fixed count, because the benchmark frontier after 2025 changes monthly. We first give a broad taxonomy of the tasks commonly used to evaluate VLMs. We then highlight the 2025–2026 shift toward embodied, temporal, and judge-aware evaluation, summarize common metric choices, and discuss the limits of current automatic evaluation practice. Our central observation is that benchmark diversity has increased faster than benchmark fidelity: many datasets broaden topics or modalities, but the hardest open problem is now whether the evaluation itself can faithfully measure grounded multimodal competence.

Benchmark Categorization. Benchmarks are designed with specific testing objectives, and we group them into a broad taxonomy of representative categories (Table 3).

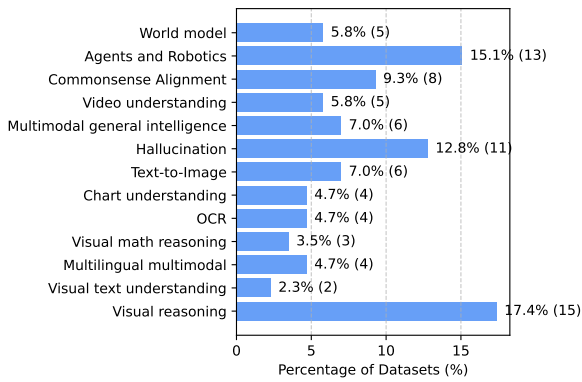
4.1. How Are Benchmark Data Collected

Benchmark datasets are typically created using one of three common data collection pipelines: fully human-annotated datasets; partially human-annotated datasets scaled up with synthetic data generation and partially validated by humans; and partially human-annotated datasets scaled up with synthetic data and fully validated by humans.

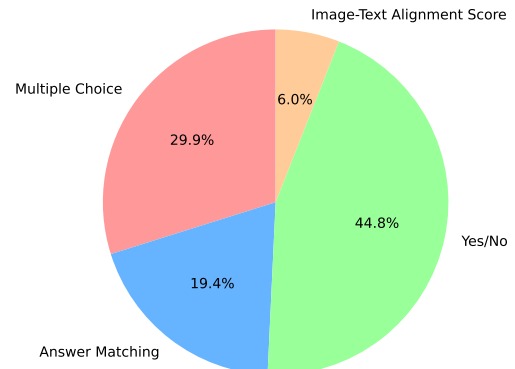
Fully human-annotated datasets are created by having humans collect or generate adversarial or challenging test questions from diverse subjects and fields. For example, MMMU[249] has 50 college students from various disciplines to collect existing test questions from textbooks and lecture materials, often in multiple choice format. Another approach involves humans creating questions and having annotators provide answers to these questions. In VCR[252], Mechanical Turks are tasked with using contexts, detected objects, and images to write one to three questions about each image, along with reasonable answers and explanations. Fully human annotated datasets are time-

Method / Work	Year	Signal	Target setting	Why it mattered
MM-RLHF [259]	2025	DPO	General multimodal preference alignment	Established large-scale human preference tuning for VLMs and critique-based reward modeling
MM-Eureka [152]	2025	RLOO [103]	Visual reasoning	Showed that rule-based reinforcement learning can induce longer and stronger multimodal reasoning traces
Vision-R1 [83]	2025	GRPO	Image reasoning	Brought DeepSeek-R1 style reasoning incentives into the VLM setting
R1-VL [256]	2025	Step-wise GRPO	Multimodal reasoning	Reinforced intermediate reasoning steps rather than only final answers
S-GRPO [236]	2026	Supervised GRPO	Cold-start post-training	Combined ground-truth trajectories with GRPO to stabilize multimodal RL from weak initial policies
Faithful GRPO [96]	2026	Constrained GRPO	Visual-spatial reasoning	Used explicit consistency constraints to reduce shortcut reasoning and answer instability
ZPPO [104]	2026	Prompt-zone policy optimization	Qwen-family VLMs	Replaced gradient-heavy teacher distillation with prompt-level teacher guidance and improved generalization
AlphaGRPO [81]	2026	GRPO + decomposed rewards	Unified multimodal generation	Extended multimodal RL beyond understanding into generation and editing

Table 2. Representative post-training methods for modern VLMs. The frontier moved from supervised instruction tuning and preference optimization toward multimodal reinforcement learning that can supervise reasoning, grounding, and generation with sparse or verifiable rewards.



(a) Most of our surveyed data tests VLMs’ visual reasoning abilities.



(b) Majority of the benchmarks are designed in multiple choice or yes/no format for ease of evaluations.

Figure 3. Our surveyed (a) benchmark dataset categories and (b) common evaluation practices. Most existing benchmarks focus on Yes/No and multiple choice format for the ease of evaluation. However, multiple choice and Yes/No questions also have their limitations that VLMs/LLMs can blindly answer above random guessing probability without giving them the questions [17]. Current scope of VLM benchmark and evaluation is broad but not comprehensive due to the challenges of reliability of answer matching.

consuming and hard to scale up, which brings inspiration to automatic question generation with human validation.

Synthetic question generation has become a more popular part of benchmark generation pipeline on various disciplines such as chart understanding [148], video understanding [147] to quickly scale up dataset sizes. Common practices include using human written examples as seed exam-

ples, giving a powerful LLM to generate more adversarial example questions and answers [107]. Often, the generation process is only involved with texts. Chart and video data are often paired with visual content and captions, which are often used by authors as context to prompt LLMs to extract answers and generate questions [114, 147]. However, LLMs are not always accurate and may produce unfaithful content

Category	Description	Datasets
Visual text understanding	Evaluates models’ ability to extract and understand texts within visual components	TextVQA [194], DocVQA [150]
Multilingual multimodal understanding	Evaluates VLMs on different languages on different tasks such as question answering and reasoning	MM-En/CN [134], CMMLU [112], C-Eval [84], MTVQA [202]
Visual math reasoning	Tests models’ ability to solve math problems in image forms	MathVista [138], MathVision [213], MM-Vet [247]
Optical Character Recognition (OCR)	Test models’ ability to extract objects from visual inputs	MM-Vet [247], OCRBench [135], MME [49], MMT-Bench [244]
Chart graphic understanding	Evaluates models’ ability to interpret graphic-related data	infographic VQA [149], AI2D [99], ChartQA [148], MMMU [249]
Text-to-Image generation	Evaluates models’ ability to generate images	MSCOCO [126], GenEval [55], T2I-CompBench [80], DPG-Bench [79], VQAScore [127], GenAI-Bench [108]
Hallucination	Evaluates whether models are likely to hallucinate on certain visual and textual inputs	HallusionBench [60], POPE [116], CHAIR [187], M-HalDetect [61], Hallu-Pi [45], Halle-Switch [253], BEAF [242], AutoHallusion [227], GAIVE [128], Hal-Eval [91], AMBER [212]
Multimodal general intelligence	Evaluates models’ ability on diverse domains of tasks	MMLU [71], MMMU [249], MMStar [32], M3GIA [197], AGIEval [263]
Video understanding	Evaluates models’ ability to understand videos (sequences of images)	EgoSchema [147], MLVU [266], MVBench [114], VideoMME [50], MovieChat [196], Perception-Test [179],
Visual reasoning, understanding, recognition, and question answering	Evaluate VLMs’ ability to recognize objects, answer questions, and reason through both visual and textual information	MMTBench [244], GQA [86], MM-En/CN [134], VCR [252], VQAv2 [58], MM-Vet [247], MMU [134], SEEDBench [107], Real World QA [229], MMMU-Pro [250], DPG [79], MSCOCO-30K [126], MM-Vet [247], ST-VQA [20], NaturalBench [109]
Alignment with common sense and physics	Evaluate the alignment between the AIGC images and videos generated by VLMs and the real world	VBench [85], PhysBench [38], VideoPhy [18], WISE [163], VideoScore [70], CRAVE [199], WorldSimBench [180], WorldModelBench [111]
GUI, web, and computer-use agents	Evaluates whether agents can ground interface elements and complete interactive tasks on screens, websites, phones, and operating systems	ScreenSpot [35], AITW [185], WebArena [267], VisualWebArena [101], OSWorld [233]
Robot benchmark and embodied control	Evaluate the embodied VLMs’ abilities online in rule-based simulators or offline datasets recording collected interactions	Habitat [189], Gibson [232], iGibson [110], Isaac Lab [155], CALVIN [151], VLMBench [262], GemBench [54], VIMA-Bench [92], VirtualHome [178], AI2-THOR [102], ProcTHOR [42], ThreeDWorld [53]
Generative model, world model	Evaluate the embodied AI models’ abilities with interactive models representing the environments	GAIA-1 [77], UniSim [238], LWM [131], Genesis [9], RoboGen [218]

Table 3. Representative benchmark categories for evaluating VLMs. Most historical benchmarks emphasize recognition, question answering, OCR, or generation quality, while newer work increasingly stresses temporal reasoning, embodied interaction, GUI/computer-use agents, and evaluation reliability in realistic settings such as web agents, robotics, and domain-specific decision making [132, 222].

or hallucinations without human supervision [59, 121, 235]. To address this, pipelines typically include automatic filters to remove low-quality outputs, followed by crowdworker validation of either randomly sampled or all generated examples [107, 147, 148]. Automatic benchmark generation helps scale dataset size with reduced human effort. However, current automatic question-generation methods primarily rely on captions and textual contexts, which can lead to the creation of questions that are easy to answer without requiring significant visual reasoning [60, 157], which undermines the benchmark’s primary goal—evaluating a VLM’s ability to comprehend and reason about visual content.

Interaction in the Simulator is mainly targeted at VLM benchmarks in robotics. It gathers data for training and evaluation by assessing the VLM-powered agents online.

As a data generation method stemming from reinforcement learning, such a data generation method is applicable for those scenarios that human-labeled datasets or synthetic datasets are hard and expensive to acquire, while the data construction follows some common rules like the physical law or some other common sense. With this rule-based data acquisition method, the outcome VLMs are more robust to the deviation within the multimodal inputs. During recent years, many works focus on realistic environments for robotics [54, 110, 151, 155, 189, 232, 262], web agents [101, 267], mobile GUI control [185], GUI grounding [35], and open-ended computer use [233]. Nonetheless, benchmarks [110, 189, 232] based on interaction traces or online execution are also widely used for VLM agent training and evaluation. Notably, more efforts have been used for generative model [238] or even world model [9, 77, 131]

Benchmark	Year	What it targets	What changed relative to older benchmarks
HumanCLAW [195]	2026	Embodied multimodal decision making	Separates a VLM’s decision quality from low-level motor execution, enabling attribution of failures to perception, planning, or actuation
GST-Bench [258]	2026	Global spatial awareness from continuous video	Requires integrating a long traversal into a coherent scene model instead of solving local, single-view perception tasks
ChronoVision [192]	2026	Multi-step temporal reasoning	Evaluates whether a model can reconstruct latent state over time instead of answering from one frame or a short clip
WorldExam [239]	2026	Reactivity of world models	Distinguishes visual plausibility from behavioral correctness by asking whether a world model reacts properly rather than merely looks realistic
OSReward [198]	2026	Computer-use reward modeling	Evaluates the judge itself for long-horizon GUI agents, acknowledging that scoring agent trajectories is now a research problem
ExtractBench [254]	2026	Enterprise document extraction	Moves document evaluation from free-form QA toward schema-grounded extraction with explicit structural targets
ConfBench [188]	2026	Confidence calibration for extraction	Measures whether the model knows when it is wrong, not only whether its extracted content matches the gold answer
RESPClinBench [19]	2026	Longitudinal clinical reasoning	Extends medical multimodal evaluation from single encounters to disease progression across time

Table 4. Representative 2025–2026 benchmark directions. The strongest recent benchmarks are less concerned with whether a model can answer another isolated visual question and more concerned with whether it can act, remember, react, and judge reliably over longer horizons.

to replace the previous simulators or datasets in generating more practical and better-quality datasets for VLMs. Though simulators are widely used in training and evaluating the VLM-power agents, the potential sim2real gap might exist when transplanting the terminal VLM into real-world applications, *i.e.* the VLM-powered agents might not be able to handle some real-world situations. More efforts towards the mitigation of these issues are still expected in this direction.

4.2. Evaluation Metrics

Benchmarks are designed for evaluation, with metrics established during their creation. VLM evaluation metrics are automatic to support repeated use at scale, and they often influence the question formats used in the benchmarks. We show the common evaluation metrics used in our surveyed benchmarks (Figure 3b, 4).

Answer matching is widely used for open-ended and closed-ended question types, where the answers are *short-form entities, long-form answers, numbers, or yes/no*. Generative VLMs are more verbose than extractive LLMs and VLMs, where they often generate verbose but correct answers [119], containment exact match [88] is a more practical version used more often in the evaluation, which includes removing articles and space of predicted answers and

check whether the normalized predicted answer is contained in the normalized gold answer [30, 106]. However, exact match tends to have high recall, which often fails to account for semantic equivalence between the gold and predicted answers, frequently misjudging human-acceptable correct answers as incorrect [26, 29, 119] and becomes impossible for benchmarks that seek long-form answers [234]. Prior to the instruction following success of LLM period, standard token overlapping scores such as F_1 , ROUGE [123], BLEU [172] to measure the similarity score between the gold and predicted answers, but start failing when generative models are generating more complex and diverse but correct answers [26, 29, 119, 234].

Thus, some of the benchmarks like MM-Vet [247] adopt LLMs to evaluate generated responses when the responses are long-form answers that requires semantic understanding to judge correctness. LLM evaluations are shown to have the highest correlations to human evaluation, but they also face the struggles of producing consistent outputs with internal model updates or changing prompt instructions [95, 146, 261]. While no current answer-matching evaluation method is perfect, *yes/no* questions are the easiest to evaluate compared to open-ended ones. As a result, most benchmarks rely on a multiple-choice format to assess VLMs (Figure 3b).

1) Answer Matching



Q: What is the price of the bananas per kg?
A: \$11.98

Q: What does the red sign say?
A: Stop

ST-VQA [20]

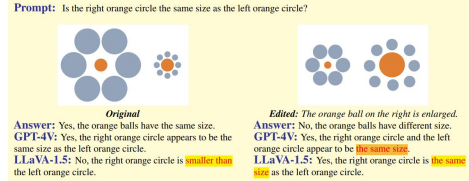
Evaluation: Accuracy
Metric: Exact Match
Format: Specific short-form answers, such as objects...



Q: Can you explain this meme?
GT: This meme is a humorous take on procrastination and the tendency to delay tasks until a specific time.

MM-Vet [245]

Evaluation: Average Score
Metric: ROUGE, LLM Eval
Format: Long-form open-ended answers

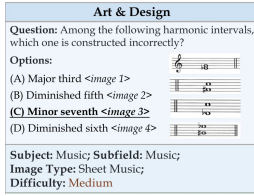


Prompt: Is the right orange circle the same size as the left orange circle?
Original
Answer: Yes, the orange balls have the same size.
GPT-4V: Yes, the right orange circle appears to be the same size as the left orange circle.
LLaVA-1.5: No, the right orange circle is smaller than the left orange circle.

HallusionBench [67]

Evaluation: Accuracy / Precision / Recall
Metric: Exact Match
Format: Yes/No question

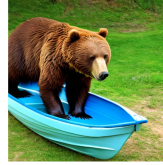
2) Multiple Choice



MMMU-Pro [248]

Evaluation: Accuracy
Metric: Exact Choice Match
Format: Multiple choice questions

3) Image-Caption Similarity



a brown bear? → 0.9925
a blue boat? → 0.9878
Score: 0.9804

"A brown bear and a blue boat"

T2I-CompBench [63]

Evaluation: Average Similarity
Metric: CLIPScore, GenEval
Format: text to image generation

Figure 4. Common benchmark evaluation metrics restrict the formats of most benchmarks, which mostly evaluate whether a VLM can generate a short-form answer that matches the correct answers.

Multiple Choice format involves selecting an answer from a set of options, including distractors, for a given visual question [107, 229, 244, 252]. This format provides definitive answers and is among the easiest to evaluate, as it measures the percentage of questions a VLM answers correctly. However, LLMs have demonstrated an unusual ability to select correct answers even without access to the actual questions [17]. Since VLMs incorporate an LLM component for generating responses (Section 3), further research is required to assess the robustness and reliability of current VLM benchmarks.

Judge reliability and calibration are emerging as benchmark targets in their own right. Recent works such as OS-Reward [198], ConfBench [188], and WorldExam [239] emphasize that for long-horizon agents, document extraction pipelines, and world models, verifying the output may be harder than generating it. This is a major departure from earlier benchmarks that assumed the scoring rule was fixed and unproblematic.

Image/text similarity scores are commonly used in image generation benchmarks like T2I-CompBench, GenEval [55, 80] to evaluate the alignment between generated images and their corresponding textual descriptions. They often rely on measures such as CLIPScore [72] for image-text alignment or ROUGE for caption matching to assess the semantic and lexical similarity between the outputs and the references.

In summary, VLM benchmarks encompass a wide range of question types, fields of expertise, and tasks, with

MMLU [71] alone covering 57 distinct tasks. However, popular evaluations remain largely confined to simple answer matching or multiple choice formats, far from the broader general intelligence of the Turing test [208].

5. Applications

Modern VLM applications are no longer limited to captioning or short-form VQA. As frontier families gained stronger grounding, longer context, and better tool use, they became practical in closed-loop settings such as screen control, web navigation, document workflows, robotics, and domain-specific decision support. These deployments also reveal the gap between benchmark performance and real-world reliability more clearly than static offline evaluation.

5.1. Computer-Use, GUI, and Web Agents

One of the clearest new application fronts is computer use: agents that read screenshots or rendered interfaces, ground language to visual elements, and emit actions such as clicks, taps, text entry, or browser navigation. ScreenAI [11] focuses on understanding user interfaces and infographics, CogAgent [75] improves high-resolution GUI grounding, and ScreenAgent [162] turns screenshot perception into a planning-acting-reflecting loop for real screen interaction. Octopus [237], ShowUI [181], and WebVoyager [66] further illustrate the move from passive screen understanding to end-to-end action on phones, desktops, and real websites.

This category matters because it stresses several capabilities at once: fine-grained OCR, element grounding, long-horizon planning, tool invocation, and recovery from execution errors. In other words, GUI and web agents turn VLMs from visual assistants into interactive operators, which is why recent benchmarks increasingly evaluate computer-use and reward-model quality rather than only answer accuracy.

5.2. Document Intelligence and Accessibility

Another rapidly growing application layer is document-centered reasoning. Interfaces, reports, dashboards, forms, and charts are often more important in practice than natural images, and they require models to combine OCR, layout understanding, numerical extraction, and grounded language generation. ScreenAI [11] is representative of this shift because it targets UI and infographic understanding directly, while ChartLLaMA [64] shows how specialized post-training can improve chart interpretation where generic models often hallucinate values or axes.

These same capabilities support accessibility-oriented applications. SciDoc2Diagrammer [157] converts scientific documents into diagrams, and recent systems for immersive or web-based accessibility use VLMs to generate contextual visual descriptions for users with low vision [167]. The practical importance of this area is high because errors are easy for end users to notice: a model that misreads a table cell, a chart label, or a form field is not merely imprecise, but directly unhelpful.

5.3. Generative Media and Design

VLMs are also increasingly used as creative tools rather than only analytical ones. In lighter-weight consumer settings, systems such as Supermeme.ai [200] use multimodal understanding and generation to produce memes and social content. At the high end, video-generation systems such as Movie Gen [177] point toward multimodal assistants that can storyboard, edit, and synthesize richer media assets from mixed text-image-video inputs.

This application category is especially relevant to the transition from Era 3a to Era 3b systems described earlier. Once a model family can both understand and generate visual content through a shared multimodal trunk, the boundary between analysis, editing, and creation becomes much thinner.

5.4. Robotics and Autonomous Systems

In physical-world settings, VLMs act as perception-and-reasoning modules for manipulation, navigation, and driving. VIMA [92] and Instruct2Act [82] show how language-conditioned visual reasoning can support manipulation, while the RT series [22, 23, 168] demonstrates the broader vision-language-action direction. For navigation, systems such as ZSON [144] and VLFM [246] use grounded multi-

modal representations to guide exploration and object-goal search. In autonomous driving, DriveVLM [205] and related systems use multimodal reasoning for scene understanding, planning, and long-tail case analysis.

These applications are among the most demanding because they convert model outputs into physical or safety-critical behavior. As a result, they expose weaknesses in calibration, latency, sim-to-real transfer, and action reliability more starkly than chatbot-style deployments do.

5.5. High-Stakes Domain Assistants

Some of the most valuable near-term uses of VLMs are expert copilots in domains where images are semantically dense but humans remain in the loop. In healthcare, VisionUnite [120] and clinician-evaluated radiology studies [243] illustrate how multimodal models can support diagnosis, explanation, and patient communication. In education, recent work uses VLMs for diagram and mathematics reasoning [223] and even for simulating students to help teachers practice instruction [142]. Related directions appear in agriculture [268], AI for science [27], and climate applications [31].

Across these domains, the pattern is consistent: VLMs are moving from passive describers to interactive copilots that must ground perception, reason over domain context, and communicate uncertainty appropriately. That shift is one reason why post-training, reward modeling, and embodied or calibration-aware evaluation have become central to modern VLM development.

6. Challenges of VLMs

This section examines key challenges in VLM research, including hallucination, safety, fairness, alignment, efficiency in training and fine-tuning, and data scarcity. Despite recent advancements, understanding these limitations is crucial to mitigating risks and ensuring ethical, reliable deployment, particularly for marginalized users.

6.1. Hallucination

Hallucination in VLMs refers to referencing nonexistent objects in images [187]. Despite benchmark-setting performance, hallucination is still a pervasive issue in VLMs, especially in visual-text tasks. Researchers have proposed datasets and metrics to quantify hallucination, with early efforts tending to require human annotation. CHAIR [187] quantifies hallucination in image captioning using per-instance and per-sentence metrics. POPE [116] assesses hallucination with Yes-No questions on object existence. M-HalDetect [61] provides 16K fine-grained VQA samples for training VLMs to detect and prevent hallucinations.

Subsequent research investigated hallucination in finer detail. Halle-Switch [253] evaluates hallucination based on data amount, quality, and granularity, balancing contextual

and parametric knowledge. Hallu-Pi [45] provides 1,260 images with detailed annotations to detect perturbation-induced hallucinations. BEAF [242] examines before-and-after image changes, introducing new metrics: true understanding, ignorance, stubbornness, and indecision. HallusionBench [60] tests VLM visual reasoning with dependent, unanswerable questions across diverse topics and formats. AutoHallusion [227] automates hallucination benchmark generation, probing VLM language modules for context-based hallucination examples.

The advent of more sophisticated LLMs has also assisted the development of larger benchmark datasets in this area. GAIVE [128] uses GPT-4 to generate 400K samples across 16 vision-language tasks, covering hallucinations like nonexistent object and knowledge manipulation. Hal-Eval [91] constructs 2M image-caption pairs, leveraging GPT-4 for fine-grained hallucination induction. AMBER [212] is an LLM-free multi-dimensional benchmark designed for generative and discriminative tasks, annotating four types of hallucination.

6.2. Safety

Given VLMs’ versatility, safeguarding against unethical use is crucial. Jailbreaking [93] allows malicious circumvention of ethical boundaries, posing risks in robotics and other downstream tasks [47, 97, 186, 226, 248]. SafeBench [245] introduces harmful queries across 23 risk scenarios using an LLM-based jury deliberation framework. MM-SafetyBench [133] evaluates VLM safety with image-text query pairs in unsafe contexts. JailbreakV [139] introduces 28K malicious image-based queries, testing attack transferability across models. SHIELD [193] evaluates face spoofing and forgery detection using True-False queries in zero- and few-shot settings. HADES [117] exploits gradient updates and adversarial methods to conceal and amplify harmful content, breaking multimodal alignment. imgJP [164] bypasses refusal guardrails using images instead of prompts, demonstrating high transferability across VLMs.

6.3. Fairness

Extensive literature has explored inequities in LLMs and VLMs [15, 52]. Like unimodal LLMs, VLMs show disparate performance, particularly affecting marginalized groups [1, 10, 87]. MMBias [89] presents a human-annotated image dataset targeting bias in religion, nationality, disability, and sexual orientation. FMBench [225] proposes a benchmark using medical images to assess gender, skin tone, and age bias. Harvard-FairVL [140] shows CLIP and BLIP2 favor Asian, Male, and Non-Hispanic groups. FairmedFM [94] integrates 17 datasets to evaluate fairness in medical tasks. CulturalVQA [161] (2,378 image-question pairs) shows better performance for North American cultures and worse performance for African and Islamic ones.

6.4. Alignment

Multi-modality Alignment. The alignment issue in multi-modal models arises from contextual deviation between modalities, leading to hallucinations [216]. Many efforts to mitigate this include leveraging VLM reasoning for self-reflection [215] or designing projectors to bridge modalities. SIMA [215] improves L-VLM alignment via self-critique and vision metrics. SAIL [257] aligns pretrained unimodal models for better multimodal learning. ExMCR [220] enables paired-data-free semantic alignment using contrastive representation. OneLLM [220] unifies eight modalities to language through a unified encoder and progressive multimodal alignment. SeeTRUE [241] benchmarks text-image alignment, proposing VQA-based and end-to-end classification methods for better misalignment detection and ranking.

Commonsense and Physics Understanding. The LVLMS used for AI-generated content (AIGC) images and videos, sometimes known as World Models [63], like SORA [24] and Veo2 [210], attract much attention throughout the community. However, these LVLMS face challenges in commonsense alignment and physics adherence. Many recent benchmarks and evaluation models aim to address these issues. VBench [85] evaluates video generative models across structured quality dimensions. PhysBench [38] and VideoPhy [18] assess VLMs and text-to-video models on physical understanding and commonsense adherence. WISE [163] introduces WiScore for T2I knowledge-image alignment. CRAVE [199] focuses on AIGC video quality assessment, aligning textual prompts with video dynamics. VideoScore [70] tracks model progress using VideoFeedback, a large-scale human-annotated dataset. WorldSimBench [180] and WorldModelBench [111] evaluate World Simulators for video-action consistency and decision-making applications. GPT4Motion [141] integrates LLMs, physics engines, and diffusion models for physics-aware text-to-video synthesis. Despite these efforts, key challenges remain in advanced video evaluation and bridging the gap between AIGC and real-world fidelity.

Training Efficiency. Efficient training and alignment of VLMs remain a very heated research topic due to their high cost and difficulty in training. Recent studies explore the impact of different pre-training settings over modules [125] or supervision [219] on the ultimate performance of VLMs. However, many applications require specialized rather than multi-task capabilities. Low-Rank Adaptation (LoRa) [44, 78] enables efficient fine-tuning with fewer parameters. RLHF [14, 105] integrates human or model feedback for alignment. Rule-based RL, requiring multiple input generations, increases computational costs, limiting its use to small VLMs [174]. Alternative RL methods (PPO, DPO) reduce computation but demand extensive human annotation to trade for computation resources [152, 259].

6.5. Data Scarcity

The abilities and reliabilities of VLMs are highly depending on the availability and diversity of the training datasets. However, the massive scale of current advanced VLMs and the scarcity of high-quality training datasets add up to the difficulty in continuously improving the performance of the future VLMs. One potential method to mitigate this issue is to use self-supervised learning (SSL) [158] that learns the representation automatically from the unlabelled dataset. Another major direction is to use the synthetic data generated by following some rules [16] or utilizing some third-party tools [191]. In VLM, specifically designed for physical world-related purposes, like robotics [201] or web agents [28], another option is to gather datasets from the interactions with the physical simulators or world model [9, 98, 218, 224], or learning from videos with human demonstrations [122, 265]. Though a lot of efforts have been made in all three directions, more insights are still expected into the breakthrough of the mass-scale training for LLMs and the alternatives to the internet-scale data, given Ilya Sutskever’s quote that “Pre-training as we know it will unquestionably end.”

7. Conclusion

Developments of VLMs and LLMs are happening at a break-neck pace with more sophisticated applications and use cases being introduced in quick succession. This paper aims to capture the most notable architectures, trends, applications along with prominent challenges in this area. We hope that our survey provides a solid general overview for practitioners as a road map for future works.

8. Limitation

Given the rapid growth of VLM research, our survey is intentionally selective rather than exhaustive. We focus on representative frontier families, benchmark directions, and post-training methods that clarify the field’s main architectural and evaluation shifts through 2026. However, the pace of releases remains faster than any static manuscript can track. New models, benchmarks, and alignment recipes will continue to appear after publication, so the companion repository and website remain the better medium for near-real-time updates.

References

- [1] Tosin Adewumi, Lama Alkhaled, Namrata Gurung, Goya van Boven, and Irene Pagliai. Fairness and bias in multimodal ai: A survey. *arXiv preprint arXiv:2406.19097*, 2024. 12
- [2] Aishwarya Agrawal, Jiasen Lu, Stanislaw Antol, Margaret Mitchell, C. Lawrence Zitnick, Dhruv Batra, and Devi Parikh. Vqa: Visual question answering, 2016. 1
- [3] Pravesh Agrawal, Szymon Antoniak, Emma Bou Hanna, Baptiste Bout, Devendra Chaplot, Jessica Chudnovsky, Diogo Costa, Baudouin De Monicault, Saurabh Garg, Theophile Gervet, Soham Ghosh, Amélie Héliou, Paul Jacob, Albert Q. Jiang, Kartik Khandelwal, Timothée Lacroix, Guillaume Lample, Diego Las Casas, Thibaut Lavril, Teven Le Scao, Andy Lo, William Marshall, Louis Martin, Arthur Mensch, Pavankumar Muddireddy, Valera Nemychnikova, Marie Pellat, Patrick Von Platen, Nikhil Raghuraman, Baptiste Rozière, Alexandre Sablayrolles, Lucile Saulnier, Romain Sauvestre, Wendy Shang, Roman Soletskyi, Lawrence Stewart, Pierre Stock, Joachim Studnia, Sandeep Subramanian, Sagar Vaze, Thomas Wang, and Sophia Yang. Pixtral 12b, 2024. 5
- [4] Moonshot AI. Kimi k3: Open frontier intelligence, 2026. Accessed: 2026-08-21. 1, 3, 5
- [5] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al. Flamingo: a visual language model for few-shot learning. *Advances in neural information processing systems*, 35: 23716–23736, 2022. 2, 5
- [6] Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, et al. Gemini: A family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 1, 2023. 3
- [7] Anthropic. Claude: An ai assistant by anthropic, 2024. Accessed: 2024-12-23. 3
- [8] Anthropic. Introducing claude opus 4.6, 2026. Accessed: 2026-08-21. 1, 3, 5
- [9] Genesis Authors. Genesis: A universal and generative physics engine for robotics and beyond, 2024. 8, 13
- [10] Rumaisa Azeem, Andrew Hundt, Masoumeh Mansouri, and Martim Brandão. Llm-driven robots risk enacting discrimination, violence, and unlawful actions. *arXiv preprint arXiv:2406.08824*, 2024. 12
- [11] Gilles Baechler, Srinivas Sunkara, Maria Wang, Fedir Zubach, Hassan Mansoor, Vincent Etter, Victor Cărbune, Jason Lin, Jindong Chen, and Abhanshu Sharma. Screenai: A vision-language model for ui and infographics understanding, 2024. 10, 11
- [12] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond, 2023. 3
- [13] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yanzhi Zhu, Mingkun Yang, Zhao-hai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report, 2025. 3
- [14] Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, et al. Training a helpful and

- harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022. 12
- [15] Zechen Bai, Pichao Wang, Tianjun Xiao, Tong He, Zongbo Han, Zheng Zhang, and Mike Zheng Shou. Hallucination of multimodal large language models: A survey. *arXiv preprint arXiv:2404.18930*, 2024. 12
- [16] Vahid Balazadeh, Mohammadmehdi Ataei, Hyunmin Cheong, Amir Hosein Khasahmadi, and Rahul G. Krishnan. Synthetic vision: Training vision-language models to understand physics, 2024. 13
- [17] Nishant Balepur, Abhilasha Ravichander, and Rachel Rudinger. Artifacts or abduction: How do LLMs answer multiple-choice questions without the question? In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10308–10330, Bangkok, Thailand, 2024. Association for Computational Linguistics. 7, 10
- [18] Hritik Bansal, Zongyu Lin, Tianyi Xie, Zeshun Zong, Michal Yarom, Yonatan Bitton, Chenfanfu Jiang, Yizhou Sun, Kai-Wei Chang, and Aditya Grover. Videophy: Evaluating physical commonsense for video generation. *arXiv preprint arXiv:2406.03520*, 2024. 8, 12
- [19] Mouxiao Bian, Zhi Chen, Ruiyao Chen, Lu Lu, Hengrui Liang, Chaoyi Huang, Yiluo Lin, Jingru Ding, et al. RESP-ClinBench: Benchmarking Multimodal Clinical Decision-Making and Longitudinal Disease Management in Respiratory Specialty Care, 2026. 9
- [20] Ali Furkan Biten, Ruben Tito, Andres Mafra, Lluís Gomez, Marçal Rusiñol, Ernest Valveny, C. V. Jawahar, and Dimosthenis Karatzas. Scene text visual question answering, 2019. 8
- [21] Florian Bordes, Richard Yuanzhe Pang, Anurag Ajay, Alexander C Li, Adrien Bardes, Suzanne Petryk, Oscar Mañas, Zhiqiu Lin, Anas Mahmoud, Bargav Jayaraman, et al. An introduction to vision-language modeling. *arXiv preprint arXiv:2405.17247*, 2024. 1
- [22] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alex Herzog, Jasmine Hsu, et al. Rt-1: Robotics transformer for real-world control at scale. *arXiv preprint arXiv:2212.06817*, 2022. 11
- [23] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023. 11
- [24] Tim Brooks, Bill Peebles, Connor Holmes, Will DePue, Yufei Guo, Li Jing, David Schnurr, Joe Taylor, Troy Luhman, Eric Luhman, et al. Video generation models as world simulators. *OpenAI Blog*, 1:8, 2024. 12
- [25] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Nee-lakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners, 2020. 5
- [26] Jannis Bulian, Christian Buck, Wojciech Gajewski, Benjamin Boerschinger, and Tal Schuster. Tomayto, tomahito. beyond token-level answer equivalence for question answering evaluation. *arXiv preprint arXiv:2202.07654*, 2022. 9
- [27] Qinglong Cao, Yuntian Chen, Lu Lu, Hao Sun, Zhenzhong Zeng, Xiaokang Yang, and Dongxiao Zhang. Promoting ai equity in science: Generalized domain prompt learning for accessible vlm research. *arXiv preprint arXiv:2405.08668*, 2024. 11
- [28] Hyunjoo Chae, Namyoung Kim, Kai Tzu iunn Ong, Minju Gwak, Gwanwoo Song, Jihoon Kim, Sunghwan Kim, Dongha Lee, and Jinyoung Yeo. Web agents with world models: Learning and leveraging environment dynamics in web navigation, 2024. 13
- [29] Anthony Chen, Gabriel Stanovsky, Sameer Singh, and Matt Gardner. Evaluating question answering evaluation. In *Proceedings of the 2nd Workshop on Machine Reading for Question Answering*, pages 119–124, Hong Kong, China, 2019. Association for Computational Linguistics. 9
- [30] Danqi Chen, Adam Fisch, Jason Weston, and Antoine Bordes. Reading wikipedia to answer open-domain questions, 2017. 9
- [31] Jian Chen, Peilin Zhou, Yining Hua, Dading Chong, Meng Cao, Yaowei Li, Zixuan Yuan, Bing Zhu, and Junwei Liang. Vision-language models meet meteorology: Developing models for extreme weather events detection with heatmaps. *arXiv preprint arXiv:2406.09838*, 2024. 11
- [32] Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, and Feng Zhao. Are we on the right way for evaluating large vision-language models?, 2024. 8
- [33] Xiaokang Chen, Zhiyu Wu, Xingchao Liu, Zizheng Pan, Wen Liu, Zhenda Xie, Xingkai Yu, and Chong Ruan. Janus-pro: Unified multimodal understanding and generation with data and model scaling, 2025. 5
- [34] Zhe Chen, Jiannan Wu, Wenhai Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zhong, Qinglong Zhang, Xizhou Zhu, Lewei Lu, Bin Li, Ping Luo, Tong Lu, Yu Qiao, and Jifeng Dai. Internvl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks, 2024. 2
- [35] Kanzhi Cheng, Qiushi Sun, Yougang Chu, Fangzhi Xu, Yantao Li, Jianbing Zhang, and Zhiyong Wu. SeeClick: Harnessing gui grounding for advanced visual gui agents, 2024. 8
- [36] Yew Ken Chia, Pengfei Hong, Lidong Bing, and Soujanya Poria. Instructeval: Towards holistic evaluation of instruction-tuned large language models, 2023. 6
- [37] Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E. Gonzalez, Ion Stoica, and Eric P. Xing. Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality, 2023. Accessed: 2024-12-23. 2

- [38] Wei Chow, Jiageng Mao, Boyi Li, Daniel Seita, Vitor Guizilini, and Yue Wang. Physbench: Benchmarking and enhancing vision-language models for physical world understanding. *arXiv preprint arXiv:2501.16411*, 2025. 8, 12
- [39] Alibaba Cloud. Alibaba unveils qwen3.8-max: Its largest and most capable flagship model to date, 2026. Accessed: 2026-08-21. 1, 3, 5
- [40] Wenliang Dai, Nayeon Lee, Boxin Wang, Zhuolin Yang, Zihan Liu, Jon Barker, Tuomas Rintamaki, Mohammad Shoeybi, Bryan Catanzaro, and Wei Ping. Nvlm: Open frontier-class multimodal llms, 2024. 2
- [41] Google DeepMind. Gemini 3.6 flash model card, 2026. Accessed: 2026-08-21. 1, 3, 5
- [42] Matt Deitke, Eli VanderBilt, Alvaro Herrasti, Luca Weihs, Kiana Ehsani, Jordi Salvador, Winson Han, Eric Kolve, Aniruddha Kembhavi, and Roozbeh Mottaghi. Proctor: Large-scale embodied ai using procedural generation. *Advances in Neural Information Processing Systems*, 35: 5982–5994, 2022. 8
- [43] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255, 2009. 2
- [44] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized llms, 2023. 12
- [45] Peng Ding, Jingyu Wu, Jun Kuang, Dan Ma, Xuezhi Cao, Xunliang Cai, Shi Chen, Jiajun Chen, and Shujian Huang. Hallu-pi: Evaluating hallucination in multi-modal large language models within perturbed inputs. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 10707–10715, 2024. 8, 12
- [46] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale, 2021. 2, 5
- [47] Jiafei Duan, Wilbert Pumacay, Nishanth Kumar, Yi Ru Wang, Shulin Tian, Wentao Yuan, Ranjay Krishna, Dieter Fox, Ajay Mandlekar, and Yijie Guo. Aha: A vision-language-model for detecting and reasoning over failures in robotic manipulation. *arXiv preprint arXiv:2410.00371*, 2024. 12
- [48] Enrico Fini*, Mustafa Shukor*, Xiujun Li, Philipp Dufter, Michal Klein, David Haldimann, Sai Aitharaju, Louis Béthune, Zhe Gan, Victor Turrisi, Alexander Toshev, Marcin Eichner, Yinfei Yang, Moin Nabi, Josh Susskind, and Alaaeldin El-Nouby*. Multimodal autoregressive pre-training of large vision encoders, 2024. 2
- [49] Chaoyou Fu, Peixian Chen, Yunhang Shen, Yulei Qin, Mengdan Zhang, Xu Lin, Jinrui Yang, Xiawu Zheng, Ke Li, Xing Sun, Yunsheng Wu, and Rongrong Ji. Mme: A comprehensive evaluation benchmark for multimodal large language models, 2024. 8
- [50] Chaoyou Fu, Yuhan Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, Peixian Chen, Yanwei Li, Shaohui Lin, Sirui Zhao, Ke Li, Tong Xu, Xiawu Zheng, Enhong Chen, Rongrong Ji, and Xing Sun. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis, 2024. 8
- [51] Chaoyou Fu, Yi-Fan Zhang, Shukang Yin, Bo Li, Xinyu Fang, Sirui Zhao, Haodong Duan, Xing Sun, Ziwei Liu, Liang Wang, Caifeng Shan, and Ran He. Mme-survey: A comprehensive survey on evaluation of multimodal llms, 2024. 6
- [52] Isabel O Gallegos, Ryan A Rossi, Joe Barrow, Md Mehrab Tanjim, Sungchul Kim, Franck Dernoncourt, Tong Yu, Ruiyi Zhang, and Nesreen K Ahmed. Bias and fairness in large language models: A survey. *Computational Linguistics*, pages 1–79, 2024. 12
- [53] Chuang Gan, Jeremy Schwartz, Seth Alter, Damian Mrowca, Martin Schrimpf, James Traer, Julian De Freitas, Jonas Kubilius, Abhishek Bhandwaldar, Nick Haber, et al. Threedworld: A platform for interactive multi-modal physical simulation. *arXiv preprint arXiv:2007.04954*, 2020. 8
- [54] Ricardo Garcia, Shizhe Chen, and Cordelia Schmid. Towards generalizable vision-language robotic manipulation: A benchmark and llm-guided 3d policy, 2024. 8
- [55] Dhruva Ghosh, Hanna Hajishirzi, and Ludwig Schmidt. Geneval: An object-focused framework for evaluating text-to-image alignment, 2023. 8, 10
- [56] Charles Goodwin and Johanne Stege Bjørndahl. Why multimodality? why co-operative action?(transcribed by j. philipsen). *Social Interaction. Video-Based Studies of Human Sociality*, 1(2), 2018. 1
- [57] Google. Gemma 4, 2026. Accessed: 2026-08-21. 1, 3, 5
- [58] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the v in vqa matter: Evaluating the role of image understanding in visual question answering, 2017. 6, 8
- [59] Feng Gu, Zongxia Li, Carlos Rafael Colon, Benjamin Evans, Ishani Mondal, and Jordan Lee Boyd-Graber. Large language models are effective human annotation assistants, but not good independent annotators, 2025. 8
- [60] Tianrui Guan, Fuxiao Liu, Xiyang Wu, Ruiqi Xian, Zongxia Li, Xiaoyu Liu, Xijun Wang, Lichang Chen, Furong Huang, Yaser Yacoob, et al. Hallusionbench: an advanced diagnostic suite for entangled language hallucination and visual illusion in large vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14375–14385, 2024. 1, 8, 12
- [61] Anisha Gunjal, Jihan Yin, and Erhan Bas. Detecting and preventing hallucinations in large vision language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 18135–18143, 2024. 8, 11
- [62] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025. 6
- [63] David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2018. 12

- [64] Yucheng Han, Chi Zhang, Xin Chen, Xu Yang, Zhibin Wang, Gang Yu, Bin Fu, and Hanwang Zhang. Chartllama: A multimodal llm for chart understanding and generation, 2023. 11
- [65] Iryna Hartsock and Ghulam Rasool. Vision-language models for medical report generation and visual question answering: A review. *Frontiers in Artificial Intelligence*, 7: 1430984, 2024. 1
- [66] Hongliang He, Wenlin Yao, Kaixin Ma, Wenhao Yu, Yong Dai, Hongming Zhang, Zhenzhong Lan, and Dong Yu. Webvoyager: Building an end-to-end web agent with large multimodal models. *arXiv preprint arXiv:2401.13919*, 2024. 10
- [67] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition, 2015. 2
- [68] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning, 2020. 5
- [69] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners, 2021. 5
- [70] Xuan He, Dongfu Jiang, Ge Zhang, Max Ku, Achint Soni, Sherman Siu, Haonan Chen, Abhuranil Chandra, Ziyang Jiang, Aaran Arulraj, et al. Videoscore: Building automatic metrics to simulate fine-grained human feedback for video generation. *arXiv preprint arXiv:2406.15252*, 2024. 8, 12
- [71] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding, 2021. 8, 10
- [72] Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. Clipscore: A reference-free evaluation metric for image captioning, 2022. 10
- [73] O.G. Holmberg, N.D. Köhler, T. Martins, et al. Self-supervised retinal thickness prediction enables deep learning from unlabelled data to boost classification of diabetic retinopathy. *Nature Machine Intelligence*, 2:719–726, 2020. 2
- [74] Haodong Hong, Sen Wang, Zi Huang, Qi Wu, and Jiajun Liu. Why only text: Empowering vision-and-language navigation with multi-modal prompts. *arXiv preprint arXiv:2406.02208*, 2024. 1
- [75] Wenyi Hong, Weihang Wang, Qingsong Lv, Jiazheng Xu, Wenmeng Yu, Junhui Ji, Yan Wang, Zihan Wang, Yuxiao Dong, Ming Ding, et al. Cogagent: A visual language model for gui agents. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14281–14290, 2024. 10
- [76] Jeremy Howard and Sebastian Ruder. Universal language model fine-tuning for text classification, 2018. 5
- [77] Anthony Hu, Lloyd Russell, Hudson Yeo, Zak Murez, George Fedoseev, Alex Kendall, Jamie Shotton, and Gianluca Corrado. Gaia-1: A generative world model for autonomous driving. *arXiv preprint arXiv:2309.17080*, 2023. 8
- [78] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021. 12
- [79] Xiwei Hu, Rui Wang, Yixiao Fang, Bin Fu, Pei Cheng, and Gang Yu. Ella: Equip diffusion models with llm for enhanced semantic alignment, 2024. 8
- [80] Kaiyi Huang, Kaiyue Sun, Enze Xie, Zhenguo Li, and Xihui Liu. T2i-compbench: A comprehensive benchmark for open-world compositional text-to-image generation, 2023. 8, 10
- [81] Runhui Huang, Jie Wu, Rui Yang, Zhe Liu, and Hengshuang Zhao. AlphaGRPO: Unlocking Self-Reflective Multimodal Generation in UMMs via Decompositional Verifiable Reward, 2026. 6, 7
- [82] Siyuan Huang, Zhengkai Jiang, Hao Dong, Yu Qiao, Peng Gao, and Hongsheng Li. Instruct2act: Mapping multimodality instructions to robotic actions with large language model. *arXiv preprint arXiv:2305.11176*, 2023. 11
- [83] Wenxuan Huang, Bohan Jia, Zijie Zhai, Shaosheng Cao, Zheyu Ye, Fei Zhao, Zhe Xu, Yao Hu, and Shaohui Lin. Vision-r1: Incentivizing reasoning capability in multimodal large language models, 2025. 6, 7
- [84] Yuzhen Huang, Yuzhuo Bai, Zhihao Zhu, Junlei Zhang, Jinghan Zhang, Tangjun Su, Junteng Liu, Chuancheng Lv, Yikai Zhang, Jiayi Lei, Yao Fu, Maosong Sun, and Junxian He. C-eval: A multi-level multi-discipline chinese evaluation suite for foundation models, 2023. 8
- [85] Ziqi Huang, Yinan He, Jiashuo Yu, Fan Zhang, Chenyang Si, Yuming Jiang, Yuanhan Zhang, Tianxing Wu, Qingyang Jin, Nattapol Chanpaisit, et al. Vbench: Comprehensive benchmark suite for video generative models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 21807–21818, 2024. 8, 12
- [86] Drew A. Hudson and Christopher D. Manning. Gqa: A new dataset for real-world visual reasoning and compositional question answering, 2019. 8
- [87] Andrew Hundt, William Agnew, Vicky Zeng, Severin Kacianka, and Matthew Gombolay. Robots enact malignant stereotypes. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, pages 743–756, 2022. 12
- [88] Gautier Izacard and Edouard Grave. Leveraging passage retrieval with generative models for open domain question answering, 2021. 9
- [89] Sepehr Janghorbani and Gerard De Melo. Multi-modal bias: Introducing a framework for stereotypical bias assessment beyond gender and race in vision–language models. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 1725–1735, 2023. 12
- [90] Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc V. Le, Yunhsuan Sung, Zhen Li, and Tom Duerig. Scaling up visual and vision-language representation learning with noisy text supervision, 2021. 2, 5
- [91] Chaoya Jiang, Hongrui Jia, Mengfan Dong, Wei Ye, Haiyang Xu, Ming Yan, Ji Zhang, and Shikun Zhang. Hal-eval: A universal and fine-grained hallucination evaluation

- framework for large vision language models. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 525–534, 2024. 8, 12
- [92] Yunfan Jiang, Agrim Gupta, Zichen Zhang, Guanzhi Wang, Yongqiang Dou, Yanjun Chen, Li Fei-Fei, Anima Anandkumar, Yuke Zhu, and Linxi Fan. Vima: General robot manipulation with multimodal prompts. *arXiv preprint arXiv:2210.03094*, 2(3):6, 2022. 8, 11
- [93] Haibo Jin, Leyang Hu, Xinuo Li, Peiyan Zhang, Chonghan Chen, Jun Zhuang, and Haohan Wang. Jailbreak-zoo: Survey, landscapes, and horizons in jailbreaking large language and vision-language models. *arXiv preprint arXiv:2407.01599*, 2024. 12
- [94] Ruinan Jin, Zikang Xu, Yuan Zhong, Qingsong Yao, Qi Dou, S Kevin Zhou, and Xiaoxiao Li. Fairmedfm: Fairness benchmarking for medical imaging foundation models. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2024. 12
- [95] Ehsan Kamalloo, Nouha Dziri, Charles Clarke, and Davood Rafiei. Evaluating open-domain question answering in the era of large language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5591–5606, Toronto, Canada, 2023. Association for Computational Linguistics. 9
- [96] Sai Srinivas Kancheti, Aditya Kanade, Rohit Sinha, Vineeth N Balasubramanian, and Tanuja Ganu. Faithful GRPO: Improving Visual Spatial Reasoning in Multimodal Language Models via Constrained Policy Optimization, 2026. 6, 7
- [97] Sathwik Karnik, Zhang-Wei Hong, Nishant Abhangi, Yen-Chen Lin, Tsun-Hsuan Wang, Christophe Dupuy, Rahul Gupta, and Pulkit Agrawal. Embodied red teaming for auditing robotic foundation models. *arXiv preprint arXiv:2411.18676*, 2024. 12
- [98] Pushkal Katara, Zhou Xian, and Katerina Fragkiadaki. Gen2sim: Scaling up robot learning in simulation with generative models. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 6672–6679. IEEE, 2024. 13
- [99] Aniruddha Kembhavi, Mike Salvato, Eric Kolve, Minjoon Seo, Hannaneh Hajishirzi, and Ali Farhadi. A diagram is worth a dozen images, 2016. 8
- [100] Wonjae Kim, Bokyung Son, and Ildoo Kim. Vilt: Vision-and-language transformer without convolution or region supervision, 2021. 2
- [101] Jing Yu Koh, Robert Lo, Lawrence Jang, Vikram Duvvur, Ming Chong Lim, Po-Yu Huang, Graham Neubig, Shuyan Zhou, Ruslan Salakhutdinov, and Daniel Fried. Visualwebarena: Evaluating multimodal agents on realistic visual web tasks. *arXiv preprint arXiv:2401.13649*, 2024. 8
- [102] Eric Kolve, Roozbeh Mottaghi, Winson Han, Eli VanderBilt, Luca Weihs, Alvaro Herrasti, Matt Deitke, Kiana Ehsani, Daniel Gordon, Yuke Zhu, et al. Ai2-thor: An interactive 3d environment for visual ai. *arXiv preprint arXiv:1712.05474*, 2017. 8
- [103] Wouter Kool, Herke van Hoof, and Max Welling. Buy 4 re-inforce samples, get a baseline for free! In *DeepRL Struct-Pred@ICLR*, 2019. 7
- [104] Byung-Kwan Lee, Ximing Lu, Shizhe Diao, Minki Kang, Saurav Muralidharan, Karan Sapra, Andrew Tao, Pavlo Molchanov, et al. Zone of Proximal Policy Optimization: Teacher in Prompts, Not Gradients, 2026. 6, 7
- [105] Harrison Lee, Samrat Phatale, Hassan Mansoor, Kellie Ren Lu, Thomas Mesnard, Johan Ferret, Colton Bishop, Ethan Hall, Victor Carbune, and Abhinav Rastogi. Rlaif: Scaling reinforcement learning from human feedback with ai feedback. 2023. 12
- [106] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks, 2021. 9
- [107] Bohao Li, Rui Wang, Guangzhi Wang, Yuying Ge, Yixiao Ge, and Ying Shan. Seed-bench: Benchmarking multimodal llms with generative comprehension, 2023. 7, 8, 10
- [108] Baiqi Li, Zhiqiu Lin, Deepak Pathak, Jiayao Li, Yixin Fei, Kewen Wu, Tiffany Ling, Xide Xia, Pengchuan Zhang, Graham Neubig, and Deva Ramanan. Genai-bench: Evaluating and improving compositional text-to-visual generation, 2024. 8
- [109] Baiqi Li, Zhiqiu Lin, Wenxuan Peng, Jean de Dieu Nyandwi, Daniel Jiang, Zixian Ma, Simran Khanuja, Ranjay Krishna, Graham Neubig, and Deva Ramanan. Naturalbench: Evaluating vision-language models on natural adversarial samples, 2024. 8
- [110] Chengshu Li, Fei Xia, Roberto Martín-Martín, Michael Lingelbach, Sanjana Srivastava, Bokui Shen, Kent Vainio, Cem Gokmen, Gokul Dharan, Tanish Jain, et al. igibson 2.0: Object-centric simulation for robot learning of everyday household tasks. *arXiv preprint arXiv:2108.03272*, 2021. 8
- [111] Dacheng Li, Yunhao Fang, Yukang Chen, Shuo Yang, Shiyi Cao, Justin Wong, Michael Luo, Xiaolong Wang, Hongxu Yin, Joseph E Gonzalez, et al. Worldmodelbench: Judging video generation models as world models. *arXiv preprint arXiv:2502.20694*, 2025. 8, 12
- [112] Haonan Li, Yixuan Zhang, Fajri Koto, Yifei Yang, Hai Zhao, Yeyun Gong, Nan Duan, and Timothy Baldwin. Cmmu: Measuring massive multitask language understanding in chinese, 2024. 8
- [113] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation, 2022. 2, 5
- [114] Kunchang Li, Yali Wang, Yanan He, Yizhuo Li, Yi Wang, Yi Liu, Zun Wang, Jilan Xu, Guo Chen, Ping Luo, Limin Wang, and Yu Qiao. Mvbench: A comprehensive multimodal video understanding benchmark, 2024. 7, 8
- [115] Liunian Harold Li, Mark Yatskar, Da Yin, Cho-Jui Hsieh, and Kai-Wei Chang. Visualbert: A simple and performant baseline for vision and language, 2019. 5

- [116] Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen. Evaluating object hallucination in large vision-language models, 2023. [8](#), [11](#)
- [117] Yifan Li, Hangyu Guo, Kun Zhou, Wayne Xin Zhao, and Ji-Rong Wen. Images are achilles’ heel of alignment: Exploiting visual vulnerabilities for jailbreaking multimodal large language models. *arXiv preprint arXiv:2403.09792*, 2024. [12](#)
- [118] Yijiang Li, Sucheng Ren, Weipeng Deng, Yuzhi Xu, Ying Gao, Edith Ngai, and Haohan Wang. Beyond finite data: Towards data-free out-of-distribution generalization via extrapolation. *arXiv preprint arXiv:2403.05523*, 2024. [1](#)
- [119] Zongxia Li, Ishani Mondal, Huy Nghiem, Yijun Liang, and Jordan Lee Boyd-Graber. PEDANTS: Cheap but effective and interpretable answer equivalence. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 9373–9398, Miami, Florida, USA, 2024. Association for Computational Linguistics. [9](#)
- [120] Zihan Li, Diping Song, Zefeng Yang, Deming Wang, Fei Li, Xiulan Zhang, Paul E Kinahan, and Yu Qiao. Visionunite: A vision-language foundation model for ophthalmology enhanced with clinical knowledge. *arXiv preprint arXiv:2408.02865*, 2024. [11](#)
- [121] Zongxia Li, Lorena Calvo-Bartolomé, Alexander Hoyle, Paiheng Xu, Alden Dima, Juan Francisco Fung, and Jordan Boyd-Graber. Large language models struggle to describe the haystack without human help: Human-in-the-loop evaluation of topic models, 2025. [8](#)
- [122] Junbang Liang, Ruoshi Liu, Ege Ozguroglu, Sruthi Sudhakar, Achal Dave, Pavel Tokmakov, Shuran Song, and Carl Vondrick. Dreamitate: Real-world visuomotor policy learning via video generation. *arXiv preprint arXiv:2406.16862*, 2024. [13](#)
- [123] Chin-Yew Lin. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain, 2004. Association for Computational Linguistics. [9](#)
- [124] Hezheng Lin, Xing Cheng, Xiangyu Wu, Fan Yang, Dong Shen, Zhongyuan Wang, Qing Song, and Wei Yuan. Cat: Cross attention in vision transformer, 2021. [2](#), [5](#)
- [125] Ji Lin, Hongxu Yin, Wei Ping, Pavlo Molchanov, Mohammad Shoeybi, and Song Han. Vila: On pre-training for visual language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 26689–26699, 2024. [12](#)
- [126] Tsung-Yi Lin, Michael Maire, Serge Belongie, Lubomir Bourdev, Ross Girshick, James Hays, Pietro Perona, Deva Ramanan, C. Lawrence Zitnick, and Piotr Dollár. Microsoft coco: Common objects in context, 2015. [8](#)
- [127] Zhiqiu Lin, Deepak Pathak, Baiqi Li, Jiayao Li, Xide Xia, Graham Neubig, Pengchuan Zhang, and Deva Ramanan. Evaluating text-to-visual generation with image-to-text generation, 2024. [8](#)
- [128] Fuxiao Liu, Kevin Lin, Linjie Li, Jianfeng Wang, Yaser Yacoob, and Lijuan Wang. Mitigating hallucination in large multi-modal models via robust instruction tuning. In *The Twelfth International Conference on Learning Representations*, 2023. [8](#), [12](#)
- [129] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning, 2023. [2](#), [5](#)
- [130] Hanchao Liu, Wenyuan Xue, Yifei Chen, Dapeng Chen, Xiutian Zhao, Ke Wang, Liping Hou, Rongjun Li, and Wei Peng. A survey on hallucination in large vision-language models, 2024. [1](#)
- [131] Hao Liu, Wilson Yan, Matei Zaharia, and Pieter Abbeel. World model on million-length video and language with blockwise ringattention, 2024. [8](#)
- [132] Mengyin Liu, Jie Jiang, Chao Zhu, and Xu-Cheng Yin. Vlpd: Context-aware pedestrian detection via vision-language semantic self-supervision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6662–6671, 2023. [8](#)
- [133] Xin Liu, Yichen Zhu, Jindong Gu, Yunshi Lan, Chao Yang, and Yu Qiao. Mm-safetybench: A benchmark for safety evaluation of multimodal large language models. In *European Conference on Computer Vision*, pages 386–403. Springer, 2024. [12](#)
- [134] Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang, Conghui He, Ziwei Liu, Kai Chen, and Dahua Lin. Mm-bench: Is your multi-modal model an all-around player?, 2024. [8](#)
- [135] Yuliang Liu, Zhang Li, Mingxin Huang, Biao Yang, Wenwen Yu, Chunyuan Li, Xu-Cheng Yin, Cheng-Lin Liu, Lianwen Jin, and Xiang Bai. Ocrbench: on the hidden mystery of ocr in large multimodal models. *Science China Information Sciences*, 67(12), 2024. [8](#)
- [136] Ziyu Liu, Zeyi Sun, Yuhang Zang, Xiaoyi Dong, Yuhang Cao, Haodong Duan, Dahua Lin, and Jiaqi Wang. Visual-rft: Visual reinforcement fine-tuning. *arXiv preprint arXiv:2503.01785*, 2025. [6](#)
- [137] Jiasen Lu, Dhruv Batra, Devi Parikh, and Stefan Lee. VILBERT: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks, 2019. [5](#)
- [138] Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts, 2024. [6](#), [8](#)
- [139] Weidi Luo, Siyuan Ma, Xiaogeng Liu, Xiaoyu Guo, and Chaowei Xiao. Jailbreakv-28k: A benchmark for assessing the robustness of multimodal large language models against jailbreak attacks. *arXiv preprint arXiv:2404.03027*, 2024. [12](#)
- [140] Yan Luo, Min Shi, Muhammad Osama Khan, Muhammad Muneeb Afzal, Hao Huang, Shuaihang Yuan, Yu Tian, Luo Song, Ava Kouhana, Tobias Elze, et al. Fairclip: Harnessing fairness in vision-language learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12289–12301, 2024. [12](#)
- [141] Jiaxi Lv, Yi Huang, Mingfu Yan, Jiancheng Huang, Jianzhuang Liu, Yifan Liu, Yafei Wen, Xiaoxin Chen, and Shifeng Chen. Gpt4motion: Scripting physical motions in text-to-video generation via blender-oriented gpt planning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1430–1440, 2024. [12](#)

- [142] Yiping Ma, Shiyu Hu, Xuchen Li, Yipei Wang, Shiqing Liu, and Kang Hao Cheong. Students rather than experts: A new ai for education pipeline to model more human-like and personalised early adolescences. *arXiv preprint arXiv:2410.15701*, 2024. 11
- [143] Muhammad Maaz, Hanoona Rasheed, Salman Khan, and Fahad Khan. Videogpt+: Integrating image and video encoders for enhanced video understanding, 2024. 2
- [144] Arjun Majumdar, Gunjan Aggarwal, Bhavika Devnani, Judy Hoffman, and Dhruv Batra. Zson: Zero-shot object-goal navigation using multimodal goal embeddings. *Advances in Neural Information Processing Systems*, 35: 32340–32352, 2022. 11
- [145] Spyros Makridakis, Fotios Petropoulos, and Yanfei Kang. Large language models: Their success and impact. *Forecasting*, 5(3):536–549, 2023. 1
- [146] Oscar Mañas, Benno Krojer, and Aishwarya Agrawal. Improving automatic vqa evaluation using large language models. *arXiv preprint arXiv:2310.02567*, 2023. 9
- [147] Karttikeya Mangalam, Raiymbek Akshulakov, and Jitendra Malik. Egoschema: A diagnostic benchmark for very long-form video language understanding, 2023. 7, 8
- [148] Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. Chartqa: A benchmark for question answering about charts with visual and logical reasoning, 2022. 7, 8
- [149] Minesh Mathew, Viraj Bagal, Rubèn Pérez Tito, Dimosthenis Karatzas, Ernest Valveny, and C. V. Jawahar. Infograph-icvqa, 2021. 8
- [150] Minesh Mathew, Dimosthenis Karatzas, and C. V. Jawahar. Docvqa: A dataset for vqa on document images, 2021. 8
- [151] Oier Mees, Lukas Hermann, Erick Rosete-Beas, and Wolfram Burgard. Calvin: A benchmark for language-conditioned policy learning for long-horizon robot manipulation tasks, 2022. 8
- [152] Fanqing Meng, Lingxiao Du, Zongkai Liu, Zhixiang Zhou, Quanfeng Lu, Daocheng Fu, Botian Shi, Wenhai Wang, Junjun He, Kaipeng Zhang, Ping Luo, Yu Qiao, Qiaosheng Zhang, and Wenqi Shao. MM-Eureka: Exploring visual aha moment with rule-based large-scale reinforcement learning. *arXiv preprint arXiv:2503.07365*, 2025. Accessed: 2025-03-12. 6, 7, 12
- [153] MiniMax. Minimax m3: Frontier coding, 1m context, native multimodality — all in one model, 2026. Accessed: 2026-08-21. 1, 3, 5
- [154] Ishan Misra and Laurens van der Maaten. Self-supervised learning of pretext-invariant representations, 2019. 5
- [155] Mayank Mittal, Calvin Yu, Qinxu Yu, Jingzhou Liu, Nikita Rudin, David Hoeller, Jia Lin Yuan, Ritvik Singh, Yunrong Guo, Hammad Mazhar, Ajay Mandlekar, Buck Babich, Gavriel State, Marco Hutter, and Animesh Garg. Orbit: A unified simulation framework for interactive robot learning environments. *IEEE Robotics and Automation Letters*, 8(6):3740–3747, 2023. 8
- [156] Atif Farid Mohammad, Bryan Clark, and Ramya Hegde. Large language model (llm) & gpt, a monolithic study in generative ai. In *2023 Congress in Computer Science, Computer Engineering, & Applied Computing (CSCE)*, pages 383–388. IEEE, 2023. 1
- [157] Ishani Mondal, Zongxia Li, Yufang Hou, Anandhavelu Natarajan, Aparna Garimella, and Jordan Lee Boyd-Graber. SciDoc2Diagrammer-MAF: Towards generation of scientific diagrams from documents guided by multi-aspect feedback refinement. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 13342–13375, Miami, Florida, USA, 2024. Association for Computational Linguistics. 8, 11
- [158] Norman Mu, Alexander Kirillov, David Wagner, and Saining Xie. Slip: Self-supervision meets language-image pre-training. In *European conference on computer vision*, pages 529–544. Springer, 2022. 13
- [159] Fionn Murtagh. Multilayer perceptrons for classification and regression. *Neurocomputing*, 2(5):183–197, 1991. 5
- [160] Sania Nayab, Giulio Rossolini, Giorgio Buttazzo, Nicola-maria Manes, and Fabrizio Giacomelli. Concise thoughts: Impact of output length on llm reasoning and cost. *arXiv preprint arXiv:2407.19825*, 2024. 1
- [161] Shravan Nayak, Kanishk Jain, Rabiul Awal, Siva Reddy, Sjoerd Steenkiste, Lisa Hendricks, Karolina Stanczak, and Aishwarya Agrawal. Benchmarking vision language models for cultural understanding. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 5769–5790, 2024. 12
- [162] Runliang Niu, Jindong Li, Shiqi Wang, Yali Fu, Xiyu Hu, Xueyuan Leng, He Kong, Yi Chang, and Qi Wang. Screenagent: A vision language model-driven computer control agent, 2024. 10
- [163] Yuwei Niu, Munan Ning, Mengren Zheng, Bin Lin, Peng Jin, Jiaqi Liao, Kunpeng Ning, Bin Zhu, and Li Yuan. Wise: A world knowledge-informed semantic evaluation for text-to-image generation, 2025. 8, 12
- [164] Zhenxing Niu, Haodong Ren, Xinbo Gao, Gang Hua, and Rong Jin. Jailbreaking attack against multimodal large language model. *arXiv preprint arXiv:2402.02309*, 2024. 12
- [165] NVIDIA, Aditi, Niket Agarwal, Arslan Ali, Jon Allen, Martin Antolini, Adeline Aubame, Alisson Azzolini, et al. Cosmos 3: Omnimodal World Models for Physical AI, 2026. 6
- [166] NVIDIA, Aarti Basant, Amlan Kar, Despoina Paschalidou, Fangyin Wei, Francesco Ferroni, Guillermo Garcia Cobo, Haithem Turki, et al. NVIDIA OmniDreams: Real-Time Generative World Model for Closed-Loop Autonomous Vehicle Simulation, 2026. 6
- [167] Elisa Ayumi Masasi de Oliveira, Diogo Fernandes Costa Silva, and Arlindo Rodrigues Galvão Filho. Improving vr accessibility through automatic 360 scene description using multimodal large language models. In *Proceedings of the 26th Symposium on Virtual and Augmented Reality*, pages 289–293, 2024. 11
- [168] Abby O'Neill, Abdul Rehman, Abhinav Gupta, Abhiram Maddukuri, Abhishek Gupta, Abhishek Padalkar, Abraham Lee, Acorn Pooley, Agrim Gupta, Ajay Mandlekar, et al. Open x-embodiment: Robotic learning datasets and rt-x models. *arXiv preprint arXiv:2310.08864*, 2023. 11

- [169] OpenAI. Gpt-4 technical report, 2024. 1, 2, 5
- [170] OpenAI. Gpt-5.6: Frontier intelligence that scales with your ambition, 2026. Accessed: 2026-08-21. 1, 3, 5
- [171] Keiron O’Shea and Ryan Nash. An introduction to convolutional neural networks, 2015. 5
- [172] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting on Association for Computational Linguistics*, page 311–318, USA, 2002. Association for Computational Linguistics. 9
- [173] Deepak Pathak, Philipp Krahenbuhl, Jeff Donahue, Trevor Darrell, and Alexei A. Efros. Context encoders: Feature learning by inpainting, 2016. 2
- [174] Yingzhe Peng, Gongrui Zhang, Miaosen Zhang, Zhiyuan You, Jie Liu, Qipeng Zhu, Kai Yang, Xingzhong Xu, Xin Geng, and Xu Yang. Lmm-rl: Empowering 3b llms with strong reasoning abilities through two-stage rule-based rl, 2025. 12
- [175] Zhiliang Peng, Wenhui Wang, Li Dong, Yaru Hao, Shaohan Huang, Shuming Ma, and Furu Wei. Kosmos-2: Grounding multimodal large language models to the world, 2023. 2
- [176] Matthew E. Peters, Sebastian Ruder, and Noah A. Smith. To tune or not to tune? adapting pretrained representations to diverse tasks, 2019. 5
- [177] Adam Polyak, Amit Zohar, Andrew Brown, Andros Tjandra, Animesh Sinha, Ann Lee, Apoorv Vyas, Bowen Shi, Chih-Yao Ma, Ching-Yao Chuang, et al. Movie gen: A cast of media foundation models. *arXiv preprint arXiv:2410.13720*, 2024. 11
- [178] Xavier Puig, Kevin Ra, Marko Boben, Jiaman Li, Tingwu Wang, Sanja Fidler, and Antonio Torralba. Virtualhome: Simulating household activities via programs. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 8494–8502, 2018. 8
- [179] Viorica Pătrăucean, Lucas Smaira, Ankush Gupta, Adrià Recasens Contintente, Larisa Markeeva, Dylan Barnase, Skanda Koppula, Joseph Heyward, Mateusz Malinowski, Yi Yang, Carl Doersch, Tatiana Matejovicova, Yury Sulsky, Antoine Miech, Alex Frechette, Hanna Klimczak, Raphael Koster, Junlin Zhang, Stephanie Winkler, Yusuf Aytar, Simon Osindero, Dima Damen, Andrew Zisserman, and João Carreira. Perception test: A diagnostic benchmark for multimodal video models, 2023. 8
- [180] Yiran Qin, Zhelun Shi, Jiwen Yu, Xijun Wang, Enshen Zhou, Lijun Li, Zhenfei Yin, Xihui Liu, Lu Sheng, Jing Shao, et al. Worldsimbench: Towards video generation models as world simulators. *arXiv preprint arXiv:2410.18072*, 2024. 8, 12
- [181] Kevin Qinghong Lin, Linjie Li, Difei Gao, Zhengyuan Yang, Shiwei Wu, Zechen Bai, Weixian Lei, Lijuan Wang, and Mike Zheng Shou. Showui: One vision-language-action model for gui visual agent. *arXiv e-prints*, pages arXiv–2411, 2024. 10
- [182] Qwen Team. Qwen3.6-35b-a3b: Agentic coding power, now open to all, 2026. Accessed: 2026-08-21. 3
- [183] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision, 2021. 2, 5
- [184] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model, 2024. 6
- [185] C. Rawles, Alice Li, Daniel Rodriguez Rodriguez, Oriana Riva, and Timothy Lillicrap. Android in the wild: A large-scale dataset for android device control, 2023. 8
- [186] Alexander Robey, Zachary Ravichandran, Vijay Kumar, Hamed Hassani, and George J Pappas. Jailbreaking llm-controlled robots. *arXiv preprint arXiv:2410.13691*, 2024. 12
- [187] Anna Rohrbach, Lisa Anne Hendricks, Kaylee Burns, Trevor Darrell, and Kate Saenko. Object hallucination in image captioning. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 4035–4045, 2018. 8, 11
- [188] Priyashree Roy, Sujitha Martin, Mohammad Rostami, Spencer Romo, Renhao Xue, Bob Strahan, Diego A. Socolinsky, Boyi Xie, and Md Mofijul Islam. Can You Trust the Confidence? ConfBench for Vision-Language Models on Document Extraction, 2026. 9, 10
- [189] Manolis Savva, Abhishek Kadian, Oleksandr Maksymets, Yili Zhao, Erik Wijmans, Bhavana Jain, Julian Straub, Jia Liu, Vladlen Koltun, Jitendra Malik, Devi Parikh, and Dhruv Batra. Habitat: A Platform for Embodied AI Research. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019. 8
- [190] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017. 6
- [191] Sahand Sharifzadeh, Christos Kaplanis, Shreya Pathak, Dharshan Kumaran, Anastasija Ilic, Jovana Mitrovic, Charles Blundell, and Andrea Banino. Synth²: Boosting visual-language models with synthetic captions and image embeddings, 2024. 13
- [192] Yifan Shen, Jian Xu, Boyi Li, Yuner Zhang, Tianjiao Yu, Bingxuan Li, Houze Yang, Rushi Wang, and Xu Cao. ChronoVision: Temporal Reasoning via Latent State Reconstruction, 2026. 9
- [193] Yichen Shi, Yuhao Gao, Yingxin Lai, Hongyang Wang, Jun Feng, Lei He, Jun Wan, Changsheng Chen, Zitong Yu, and Xiaochun Cao. Shield: An evaluation benchmark for face spoofing and forgery detection with multimodal large language models. *arXiv preprint arXiv:2402.04178*, 2024. 12
- [194] Amanpreet Singh, Vivek Natarajan, Meet Shah, Yu Jiang, Xinlei Chen, Dhruv Batra, Devi Parikh, and Marcus Rohrbach. Towards vqa models that can read, 2019. 8
- [195] Li Siyao, Jiawei Gu, Shuai Liu, Kairui Hu, Zekun Li, Linjie Li, Chengcheng Tang, Po-Chen Wu, et al. HumanCLAW: Can Vision-Language Models Act Through a Body?, 2026. 9
- [196] Enxin Song, Wenhao Chai, Guan hong Wang, Yucheng Zhang, Haoyang Zhou, Feiyang Wu, Haozhe Chi, Xun Guo, Tian Ye, Yanting Zhang, Yan Lu, Jenq-Neng Hwang,

- and Gaoang Wang. Moviechat: From dense token to sparse memory for long video understanding, 2024. 8
- [197] Wei Song, Yadong Li, Jianhua Xu, Guowei Wu, Lingfeng Ming, Kexin Yi, Weihua Luo, Houyi Li, Yi Du, Fangda Guo, and Kaicheng Yu. M3gia: A cognition inspired multilingual and multimodal general intelligence ability benchmark, 2024. 8
- [198] Qiushi Sun, Kanzhi Cheng, Yian Wang, Bowen Yang, Hang Yan, Liheng Chen, Fangzhi Xu, Zichen Ding, et al. OS-Reward: Instituting Standardized Evaluation for Cross-Platform Computer-Use Reward Models, 2026. 9, 10
- [199] Shangkun Sun, Xiaoyu Liang, Bowen Qu, and Wei Gao. Content-rich aigc video quality assessment via intricate text alignment and motion-aware consistency. *arXiv preprint arXiv:2502.04076*, 2025. 8, 12
- [200] Supermeme.ai. Supermeme.ai: Turn text into memes using ai. <https://supermeme.ai/>, 2024. Accessed: 2024-12-24. 11
- [201] Grace Tang, Swetha Rajkumar, Yifei Zhou, Homer Rich Walke, Sergey Levine, and Kuan Fang. Kalie: Fine-tuning vision-language models for open-world manipulation without robot data. *arXiv preprint arXiv:2409.14066*, 2024. 13
- [202] Jingqun Tang, Qi Liu, Yongjie Ye, Jinghui Lu, Shu Wei, Chunhui Lin, Wanqing Li, Mohamad Fitri Faiz Bin Mahmood, Hao Feng, Zhen Zhao, Yanjie Wang, Yuliang Liu, Hao Liu, Xiang Bai, and Can Huang. Mtvqa: Benchmarking multilingual text-centric visual question answering, 2024. 8
- [203] DreamX Team, Yancheng Bai, Rui Chen, Xiangxiang Chu, Rujing Dang, Hao Dou, Bingjie Gao, Qiwen Gu, et al. DreamX-World 1.0: A General-Purpose Interactive World Model, 2026. 6
- [204] Kairos Team, Fei Wang, Shan You, Qiming Zhang, Tao Huang, Zuoyi Fu, Zhisheng Zheng, Yunlong Xi, et al. Kairos: A Regret-Aware Native World-Action Model Stack for Physical AI, 2026. 6
- [205] Xiaoyu Tian, Junru Gu, Bailin Li, Yicheng Liu, Yang Wang, Zhiyong Zhao, Kun Zhan, Peng Jia, Xianpeng Lang, and Hang Zhao. Drivevlm: The convergence of autonomous driving and large vision-language models, 2024. 1, 11
- [206] Hugo Touvron, Louis Martin, and et al. Llama 2: Open foundation and fine-tuned chat models, 2023. 1, 2, 5
- [207] Maria Tsimpoukelli, Jacob Menick, Serkan Cabi, S. M. Ali Eslami, Oriol Vinyals, and Felix Hill. Multimodal few-shot learning with frozen language models, 2021. 5
- [208] A. M. Turing. Computing machinery and intelligence. *Mind*, 59(236):433–460, 1950. 10
- [209] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding, 2019. 5
- [210] Veo-Team. Veo 2. 2024. 12
- [211] Pablo Villalobos, Anson Ho, Jaime Sevilla, Tamay Besiroglu, Lennart Heim, and Marius Hobbhahn. Position: Will we run out of data? limits of llm scaling based on human-generated data. In *Forty-first International Conference on Machine Learning*, 2024. 1
- [212] Junyang Wang, Yuhang Wang, Guohai Xu, Jing Zhang, Yukai Gu, Haitao Jia, Ming Yan, Ji Zhang, and Jitao Sang. An llm-free multi-dimensional benchmark for mllms hallucination evaluation. *arXiv preprint arXiv:2311.07397*, 2023. 8, 12
- [213] Ke Wang, Juntong Pan, Weikang Shi, Zimu Lu, Mingjie Zhan, and Hongsheng Li. Measuring multimodal mathematical reasoning with math-vision dataset, 2024. 8
- [214] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Yang Fan, Kai Dang, Mengfei Du, Xuancheng Ren, Rui Men, Dayiheng Liu, Chang Zhou, Jingren Zhou, and Junyang Lin. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution, 2024. 2, 5
- [215] Xiyao Wang, Jiahai Chen, Zhaoyang Wang, Yuhang Zhou, Yiyang Zhou, Huaxiu Yao, Tianyi Zhou, Tom Goldstein, Parminder Bhatia, Furong Huang, et al. Enhancing visual-language modality alignment in large vision language models via self-improvement. *arXiv preprint arXiv:2405.15973*, 2024. 12
- [216] Xintong Wang, Jingheng Pan, Liang Ding, and Chris Bie-mann. Mitigating hallucinations in large vision-language models with instruction contrastive decoding. *arXiv preprint arXiv:2403.18715*, 2024. 12
- [217] Xinlong Wang, Xiaosong Zhang, Zhengxiong Luo, Quan Sun, Yufeng Cui, Jinsheng Wang, Fan Zhang, Yueze Wang, Zhen Li, Qiyang Yu, Yingli Zhao, Yulong Ao, Xuebin Min, Tao Li, Boya Wu, Bo Zhao, Bowen Zhang, Liangdong Wang, Guang Liu, Zheqi He, Xi Yang, Jingjing Liu, Yonghua Lin, Tiejun Huang, and Zhongyuan Wang. Emu3: Next-token prediction is all you need, 2024. 5
- [218] Yufei Wang, Zhou Xian, Feng Chen, Tsun-Hsuan Wang, Yian Wang, Katerina Fragkiadaki, Zackory Erickson, David Held, and Chuang Gan. Robogen: Towards unleashing infinite data for automated robot learning via generative simulation. *arXiv preprint arXiv:2311.01455*, 2023. 8, 13
- [219] Zirui Wang, Jiahui Yu, Adams Wei Yu, Zihang Dai, Yulia Tsvetkov, and Yuan Cao. Simvlm: Simple visual language model pretraining with weak supervision, 2021. 12
- [220] Zehan Wang, Ziang Zhang, Luping Liu, Yang Zhao, Haifeng Huang, Tao Jin, and Zhou Zhao. Extending multi-modal contrastive representations. *arXiv preprint arXiv:2310.08884*, 2023. 12
- [221] Haoran Wei, Yaofeng Sun, and Yukun Li. Deepseek-ocr 2: Visual causal flow, 2026. 3
- [222] Dongming Wu, Wencheng Han, Tiancai Wang, Xingping Dong, Xiangyu Zhang, and Jianbing Shen. Referring multi-object tracking. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 14633–14642, 2023. 8
- [223] Di Wu, Meng Chen, Xu Chen, and Xing Liu. Analyzing k-12 ai education: A large language model study of classroom instruction on learning theories, pedagogy, tools, and ai literacy. *Computers and Education: Artificial Intelligence*, 7: 100295, 2024. 11
- [224] Philipp Wu, Alejandro Escontrela, Danijar Hafner, Pieter Abbeel, and Ken Goldberg. Daydreamer: World models for

- physical robot learning. In *Conference on robot learning*, pages 2226–2240. PMLR, 2023. [13](#)
- [225] Peiran Wu, Che Liu, Canyu Chen, Jun Li, Cosmin I Bercea, and Rossella Arcucci. Fmbench: Benchmarking fairness in multimodal large language models on medical tasks. *arXiv preprint arXiv:2410.01089*, 2024. [12](#)
- [226] Xiyang Wu, Souradip Chakraborty, Ruiqi Xian, Jing Liang, Tianrui Guan, Fuxiao Liu, Brian M. Sadler, Dinesh Manocha, and Amrit Singh Bedi. Highlighting the safety concerns of deploying llms/vlms in robotics, 2024. [12](#)
- [227] Xiyang Wu, Tianrui Guan, Dianqi Li, Shuaiyi Huang, Xiaoyu Liu, Xijun Wang, Ruiqi Xian, Abhinav Shrivastava, Furong Huang, Jordan Lee Boyd-Graber, Tianyi Zhou, and Dinesh Manocha. Autohallusion: Automatic generation of hallucination benchmarks for vision-language models, 2024. [8](#), [12](#)
- [228] Zhiyu Wu, Xiaokang Chen, Zizheng Pan, Xingchao Liu, Wen Liu, Damai Dai, Huazuo Gao, Yiyang Ma, Chengyue Wu, Bingxuan Wang, Zhenda Xie, Yu Wu, Kai Hu, Jiawei Wang, Yaofeng Sun, Yukun Li, Yishi Piao, Kang Guan, Aixin Liu, Xin Xie, Yuxiang You, Kai Dong, Xingkai Yu, Haowei Zhang, Liang Zhao, Yisong Wang, and Chong Ruan. Deepseek-vl2: Mixture-of-experts vision-language models for advanced multimodal understanding, 2024. [1](#), [3](#), [5](#)
- [229] X.AI. Grok-1.5v: The multimodal version of our ai. Blog post, 2024. Accessed on [Insert Date of Access]. [8](#), [10](#)
- [230] xAI. Introducing grok 4.5, 2026. Accessed: 2026-08-21. [1](#), [3](#), [5](#)
- [231] xAI. Grok 4.5 model reference, 2026. Accessed: 2026-08-21. [3](#)
- [232] Fei Xia, Amir R Zamir, Zhiyang He, Alexander Sax, Jitendra Malik, and Silvio Savarese. Gibson env: Real-world perception for embodied agents. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 9068–9079, 2018. [8](#)
- [233] Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, Yitao Liu, Yiheng Xu, Shuyan Zhou, Silvio Savarese, Caiming Xiong, Victor Zhong, and Tao Yu. Osvworld: Benchmarking multimodal agents for open-ended tasks in real computer environments, 2024. [8](#)
- [234] Fangyuan Xu, Yixiao Song, Mohit Iyyer, and Eunsol Choi. A critical evaluation of evaluations for long-form question answering. *ArXiv*, abs/2305.18201, 2023. [9](#)
- [235] Ziwei Xu, Sanjay Jain, and Mohan Kankanhalli. Hallucination is inevitable: An innate limitation of large language models, 2024. [8](#)
- [236] Yuming Yan, Kai Tang, Sihong Chen, Ke Xu, Dan Hu, Qun Yu, and Pengfei Hu. S-GRPO: Unified Post-Training for Large Vision-Language Models, 2026. [6](#), [7](#)
- [237] Jingkan Yang, Yuhao Dong, Shuai Liu, Bo Li, Ziyue Wang, Chencheng Jiang, Haoran Tan, Jiamu Kang, Yuanhan Zhang, Kaiyang Zhou, and Ziwei Liu. Octopus: Embodied vision-language programmer from environmental feedback, 2024. [10](#)
- [238] Mengjiao Yang, Yilun Du, Kamyar Ghasemipour, Jonathan Tompson, Dale Schuurmans, and Pieter Abbeel. Learning interactive real-world simulators. *arXiv preprint arXiv:2310.06114*, 2023. [8](#)
- [239] Yuxue Yang, Shuyao Shang, Jiahe Wang, Zitong Zhou, Liang Tan, Junhan Zeng, Ruizhi Li, Junyan Li, et al. WorldExam: Benchmarking World Models from Apparent Appearance to Inherent Reactivity, 2026. [9](#), [10](#)
- [240] Zhengyuan Yang, Linjie Li, Kevin Lin, Jianfeng Wang, Chung-Ching Lin, Zicheng Liu, and Lijuan Wang. The dawn of lmms: Preliminary explorations with gpt-4v (ision). *arXiv preprint arXiv:2309.17421*, 9(1):1, 2023. [3](#), [5](#)
- [241] Michal Yarom, Yonatan Bitton, Soravit Changpinyo, Roei Aharoni, Jonathan Herzig, Oran Lang, Eran Ofek, and Idan Szepkter. What you see is what you read? improving text-image alignment evaluation. *Advances in Neural Information Processing Systems*, 36:1601–1619, 2023. [12](#)
- [242] Moon Ye-Bin, Nam Hyeon-Woo, Wonseok Choi, and Tae-Hyun Oh. Beaf: Observing before-after changes to evaluate hallucination in vision-language models. In *European Conference on Computer Vision*, pages 232–248. Springer, 2025. [8](#), [12](#)
- [243] Nur Yildirim, Hannah Richardson, Maria Teodora Wetscherek, Junaid Bajwa, Joseph Jacob, Mark Ames Pinnock, Stephen Harris, Daniel Coelho De Castro, Shruthi Bannur, Stephanie Hyland, et al. Multimodal healthcare ai: identifying and designing clinically relevant vision-language applications for radiology. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, pages 1–22, 2024. [11](#)
- [244] Kaining Ying, Fanqing Meng, Jin Wang, Zhiqian Li, Han Lin, Yue Yang, Hao Zhang, Wenbo Zhang, Yuqi Lin, Shuo Liu, Jiayi Lei, Quanfeng Lu, Runjian Chen, Peng Xu, Renrui Zhang, Haozhe Zhang, Peng Gao, Yali Wang, Yu Qiao, Ping Luo, Kaipeng Zhang, and Wenqi Shao. Mmt-bench: A comprehensive multimodal benchmark for evaluating large vision-language models towards multitask agi, 2024. [8](#), [10](#)
- [245] Zonghao Ying, Aishan Liu, Siyuan Liang, Lei Huang, Jinyang Guo, Wenbo Zhou, Xianglong Liu, and Dacheng Tao. Safebench: A safety evaluation framework for multimodal large language models. *arXiv preprint arXiv:2410.18927*, 2024. [12](#)
- [246] Naoki Yokoyama, Sehoon Ha, Dhruv Batra, Jiuguang Wang, and Bernadette Bucher. Vlfm: Vision-language frontier maps for zero-shot semantic navigation. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 42–48. IEEE, 2024. [11](#)
- [247] Weihao Yu, Zhengyuan Yang, Linjie Li, Jianfeng Wang, Kevin Lin, Zicheng Liu, Xinchao Wang, and Lijuan Wang. Mm-vet: Evaluating large multimodal models for integrated capabilities, 2024. [8](#), [9](#)
- [248] Lu Yue, Dongliang Zhou, Liang Xie, Feitian Zhang, Ye Yan, and Erwei Yin. Safe-vln: Collision avoidance for vision-and-language navigation of autonomous robots operating in continuous environments. *IEEE Robotics and Automation Letters*, 2024. [12](#)

- [249] Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, Cong Wei, Botao Yu, Ruibin Yuan, Renliang Sun, Ming Yin, Boyuan Zheng, Zhenzhu Yang, Yibo Liu, Wenhao Huang, Huan Sun, Yu Su, and Wenhao Chen. Mmmu: A massive multi-discipline multi-modal understanding and reasoning benchmark for expert agi, 2024. [6](#), [8](#)
- [250] Xiang Yue, Tianyu Zheng, Yuansheng Ni, Yubo Wang, Kai Zhang, Shengbang Tong, Yuxuan Sun, Botao Yu, Ge Zhang, Huan Sun, Yu Su, Wenhao Chen, and Graham Neubig. Mmmu-pro: A more robust multi-discipline multi-modal understanding benchmark, 2024. [8](#)
- [251] Amir Zamir, Alexander Sax, William Shen, Leonidas Guibas, Jitendra Malik, and Silvio Savarese. Taskonomy: Disentangling task transfer learning, 2018. [2](#)
- [252] Rowan Zellers, Yonatan Bisk, Ali Farhadi, and Yejin Choi. From recognition to cognition: Visual commonsense reasoning, 2019. [6](#), [8](#), [10](#)
- [253] Bohan Zhai, Shijia Yang, Xiangchen Zhao, Chenfeng Xu, Sheng Shen, Dongdi Zhao, Kurt Keutzer, Manling Li, Tan Yan, and Xiangjun Fan. Halle-switch: Rethinking and controlling object existence hallucinations in large vision language models for detailed caption. *arXiv preprint arXiv:2310.01779*, 2023. [8](#), [11](#)
- [254] Boyang Zhang, Adrian Lyjak, Eli Stewart, Zhaoqi Li, and Simon Suo. ExtractBench: A Benchmark for Schema-Guided Enterprise Document Extraction, 2026. [9](#)
- [255] Jingyi Zhang, Jiaying Huang, Sheng Jin, and Shijian Lu. Vision-language models for vision tasks: A survey, 2024. [6](#)
- [256] Jingyi Zhang, Jiaying Huang, Huanjin Yao, Shunyu Liu, Xikun Zhang, Shijian Lu, and Dacheng Tao. R1-vl: Learning to reason with multimodal large language models via step-wise group relative policy optimization, 2025. [6](#), [7](#)
- [257] Le Zhang, Qian Yang, and Aishwarya Agrawal. Assessing and learning alignment of unimodal vision and language models, 2024. [12](#)
- [258] Qifeng Zhang, Kaixiang Huang, Heng Dong, Huang Fang, Junting Chen, Junjie Zhu, Yonghang Chen, Zhiyu Zhang, and Wei Li. GST-Bench: Can VLMs Develop Global Spatial Awareness from Video?, 2026. [9](#)
- [259] Yi-Fan Zhang, Tao Yu, Haochen Tian, Chaoyou Fu, Peiyan Li, Jianshu Zeng, Wulin Xie, Yang Shi, Huanyu Zhang, Junkang Wu, et al. Mm-rlhf: The next step forward in multimodal llm alignment. *arXiv preprint arXiv:2502.10391*, 2025. [6](#), [7](#), [12](#)
- [260] Long Zhao, Nitesh B. Gundavarapu, Liangzhe Yuan, Hao Zhou, Shen Yan, Jennifer J. Sun, Luke Friedman, Rui Qian, Tobias Weyand, Yue Zhao, Rachel Hornung, Florian Schroff, Ming-Hsuan Yang, David A. Ross, Huisheng Wang, Hartwig Adam, Mikhail Sirotenko, Ting Liu, and Boqing Gong. Videoprism: A foundational visual encoder for video understanding, 2024. [2](#)
- [261] Yilun Zhao, Haowei Zhang, Shengyun Si, Linyong Nan, Xiangru Tang, and Arman Cohan. Large language models are effective table-to-text generators, evaluators, and feedback providers. *arXiv preprint arXiv:2305.14987*, 2023. [9](#)
- [262] Kaizhi Zheng, Xiaotong Chen, Odest Jenkins, and Xin Eric Wang. VLMbench: A compositional benchmark for vision-and-language manipulation. In *Thirty-sixth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2022. [8](#)
- [263] Wanjun Zhong, Ruixiang Cui, Yiduo Guo, Yaobo Liang, Shuai Lu, Yanlin Wang, Amin Saied, Weizhu Chen, and Nan Duan. Agieval: A human-centric benchmark for evaluating foundation models, 2023. [8](#)
- [264] Hengguang Zhou, Xirui Li, Ruochen Wang, Minhao Cheng, Tianyi Zhou, and Cho-Jui Hsieh. R1-zero’s” aha moment” in visual reasoning on a 2b non-sft model. *arXiv preprint arXiv:2503.05132*, 2025. [6](#)
- [265] Huayi Zhou, Ruixiang Wang, Yunxin Tai, Yueci Deng, Guiliang Liu, and Kui Jia. You only teach once: Learn one-shot bimanual robotic manipulation from video demonstrations. *arXiv preprint arXiv:2501.14208*, 2025. [13](#)
- [266] Junjie Zhou, Yan Shu, Bo Zhao, Boya Wu, Shitao Xiao, Xi Yang, Yongping Xiong, Bo Zhang, Tiejun Huang, and Zheng Liu. Mlvu: A comprehensive benchmark for multi-task long video understanding, 2024. [8](#)
- [267] Shuyan Zhou, Frank F Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Yonatan Bisk, Daniel Fried, Uri Alon, et al. Webarena: A realistic web environment for building autonomous agents. *arXiv preprint arXiv:2307.13854*, 2023. [8](#)
- [268] Hongyan Zhu, Shuai Qin, Min Su, Chengzhi Lin, Anjie Li, and Junfeng Gao. Harnessing large vision and language models in agriculture: A review. *arXiv preprint arXiv:2407.19679*, 2024. [11](#)